



Digital Integrated Circuits

A Design Perspective

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Semiconductor Memories

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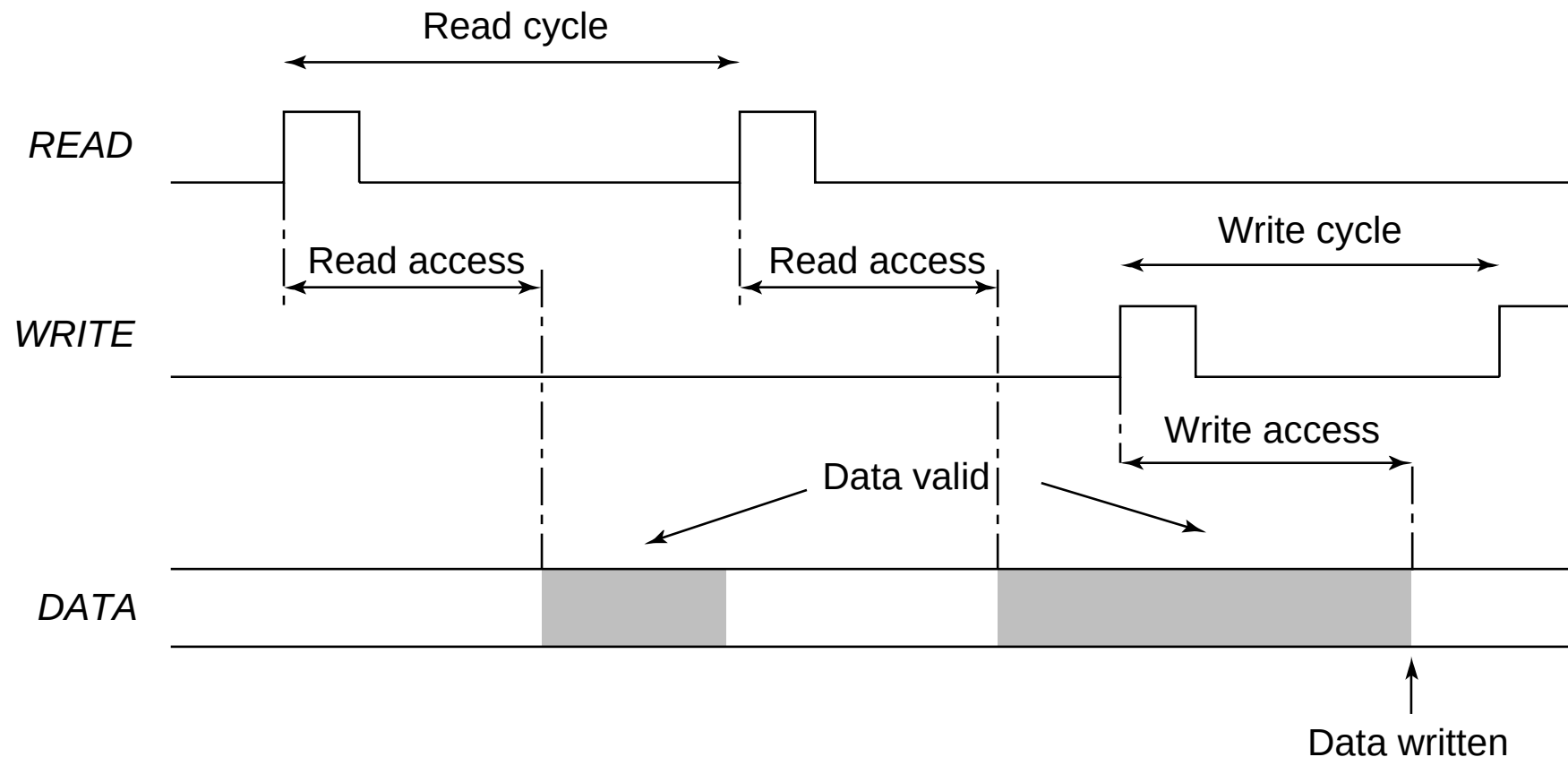
Chapter Overview

- ☐ **Memory Classification**
- ☐ **Memory Architectures**
- ☐ **The Memory Core**
- ☐ **Periphery**
- ☐ **Reliability**
- ☐ **Case Studies**

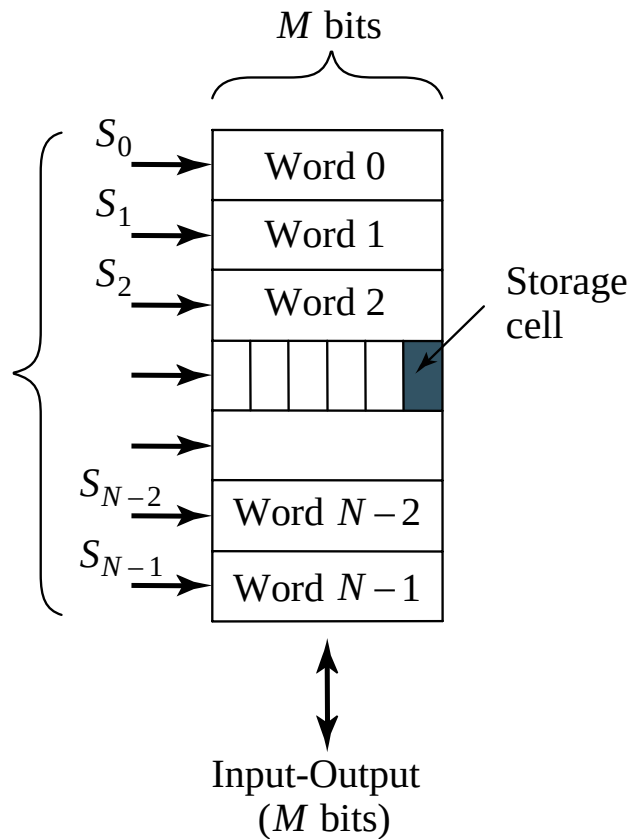
Semiconductor Memory Classification

Read-Write Memory		Non-Volatile Read-Write Memory	Read-Only Memory
Random Access	Non-Random Access	EPROM E ² PROM FLASH	Mask-Programmed Programmable (PROM)
SRAM DRAM	FIFO LIFO Shift Register CAM		

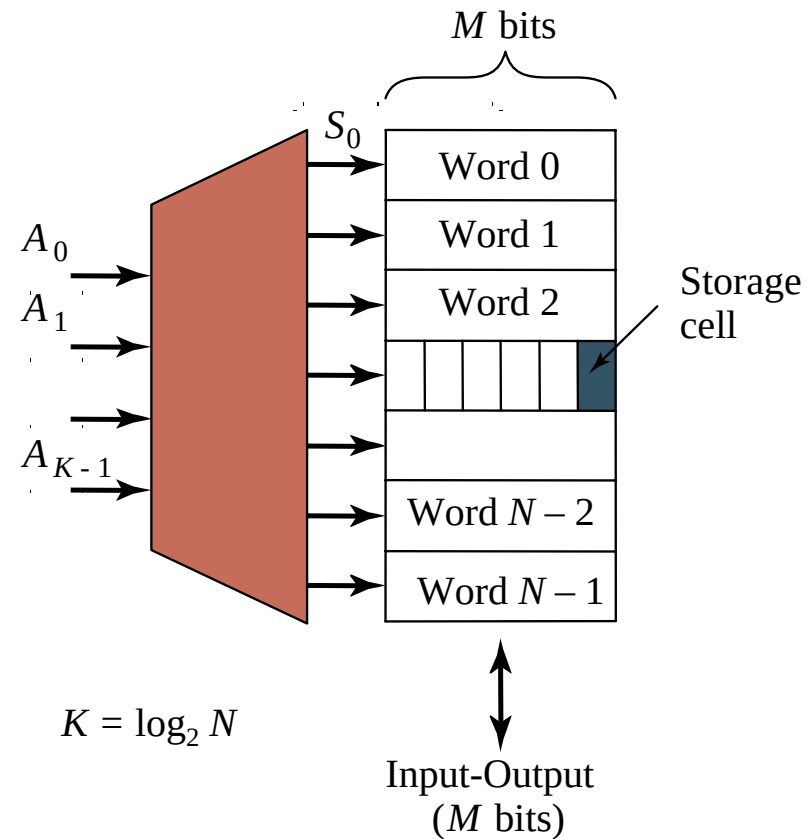
Memory Timing: Definitions



Memory Architecture: Decoders



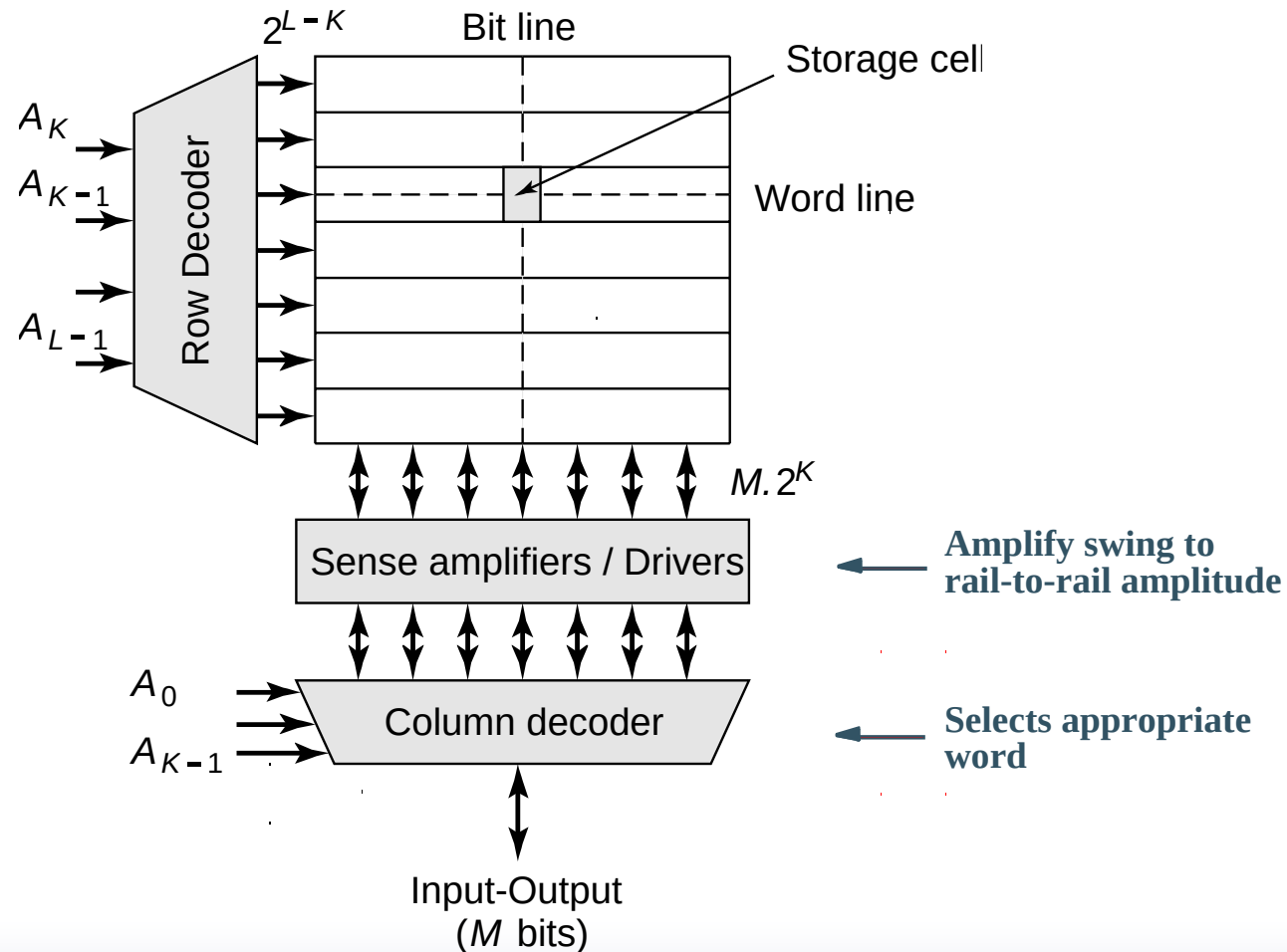
Intuitive architecture for $N \times M$ memory
Too many select signals:
 N words == N select signals



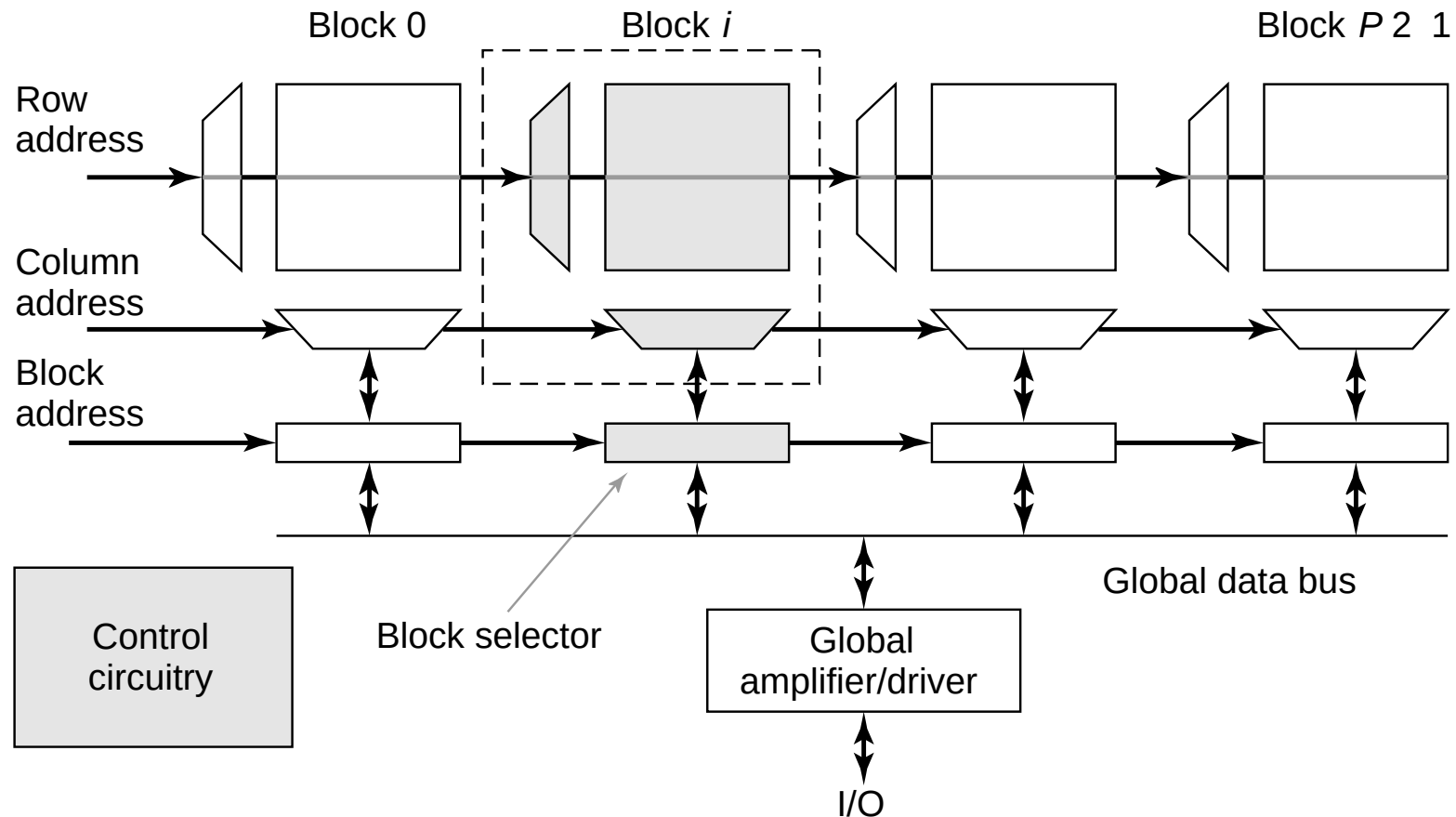
Decoder reduces the number of select signals
 $K = \log_2 N$

Array-Structured Memory Architecture

Problem: ASPECT RATIO or HEIGHT $\gg$ WIDTH



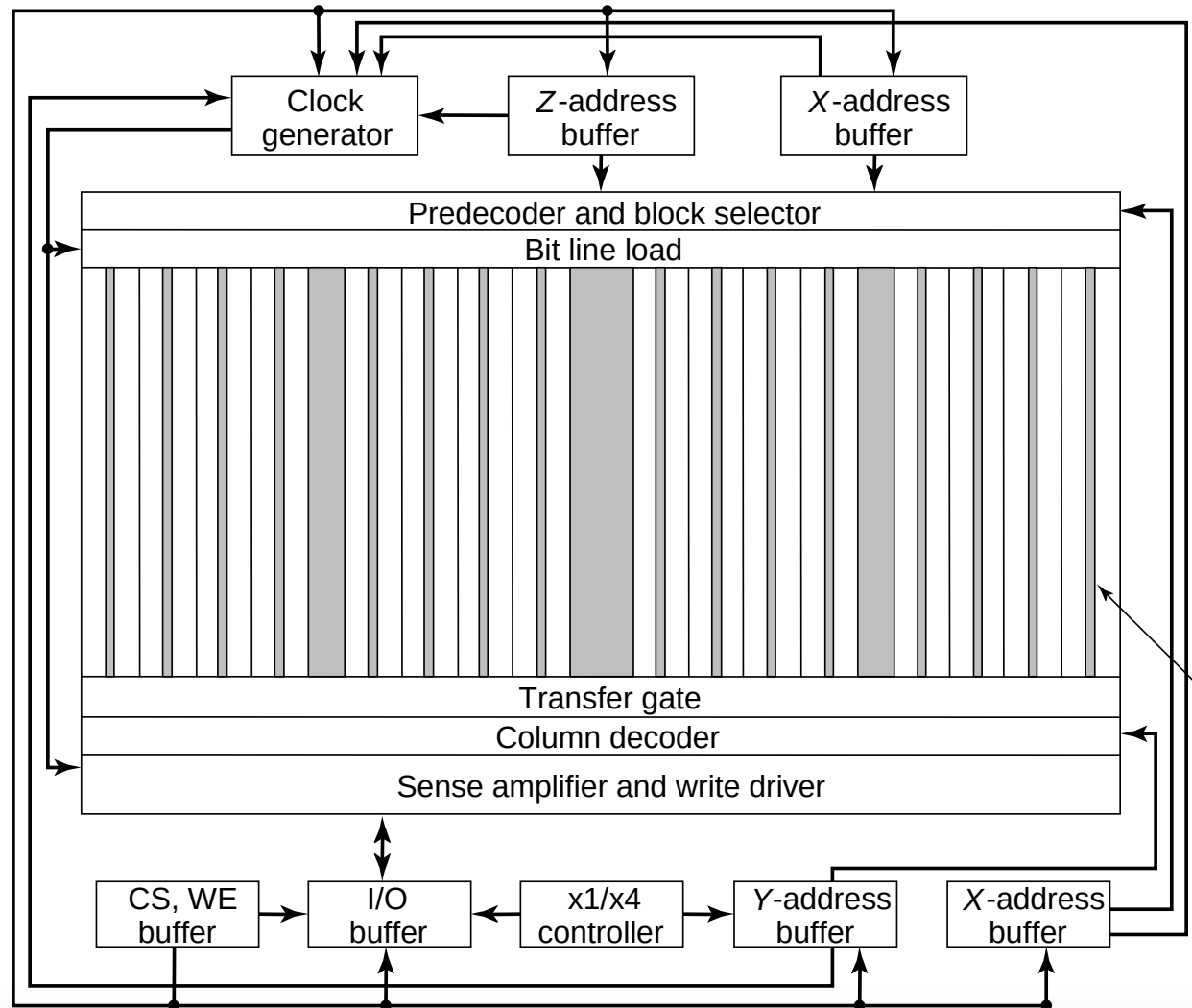
Hierarchical Memory Architecture



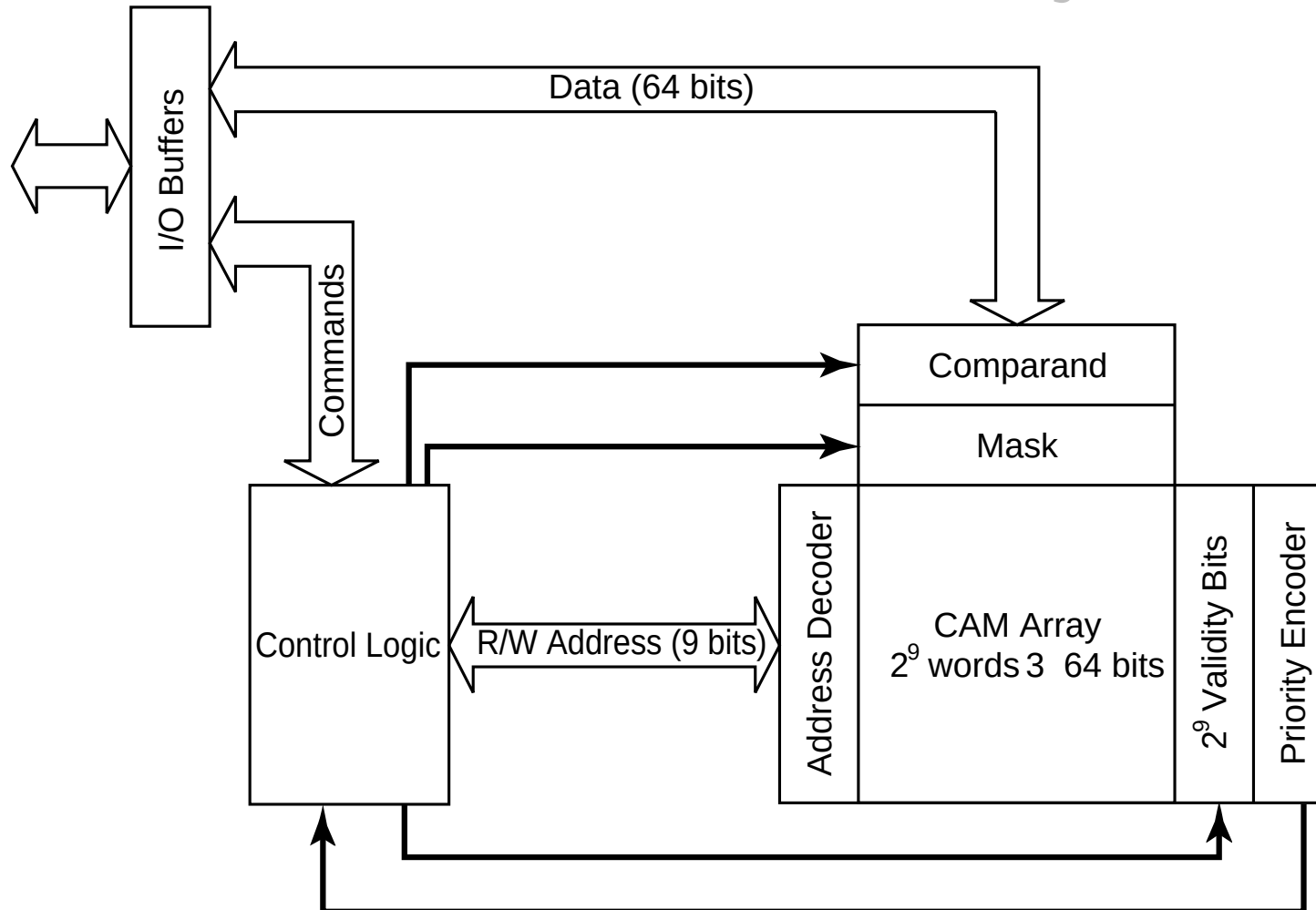
Advantages:

1. Shorter wires within blocks
2. Block address activates only 1 block => power savings

Block Diagram of 4 Mbit SRAM



Contents-Addressable Memory



Read-Write Memories (RAM)

❑ STATIC (SRAM)

Data stored as long as supply is applied

Large (6 transistors/cell)

Fast

Differential

❑ DYNAMIC (DRAM)

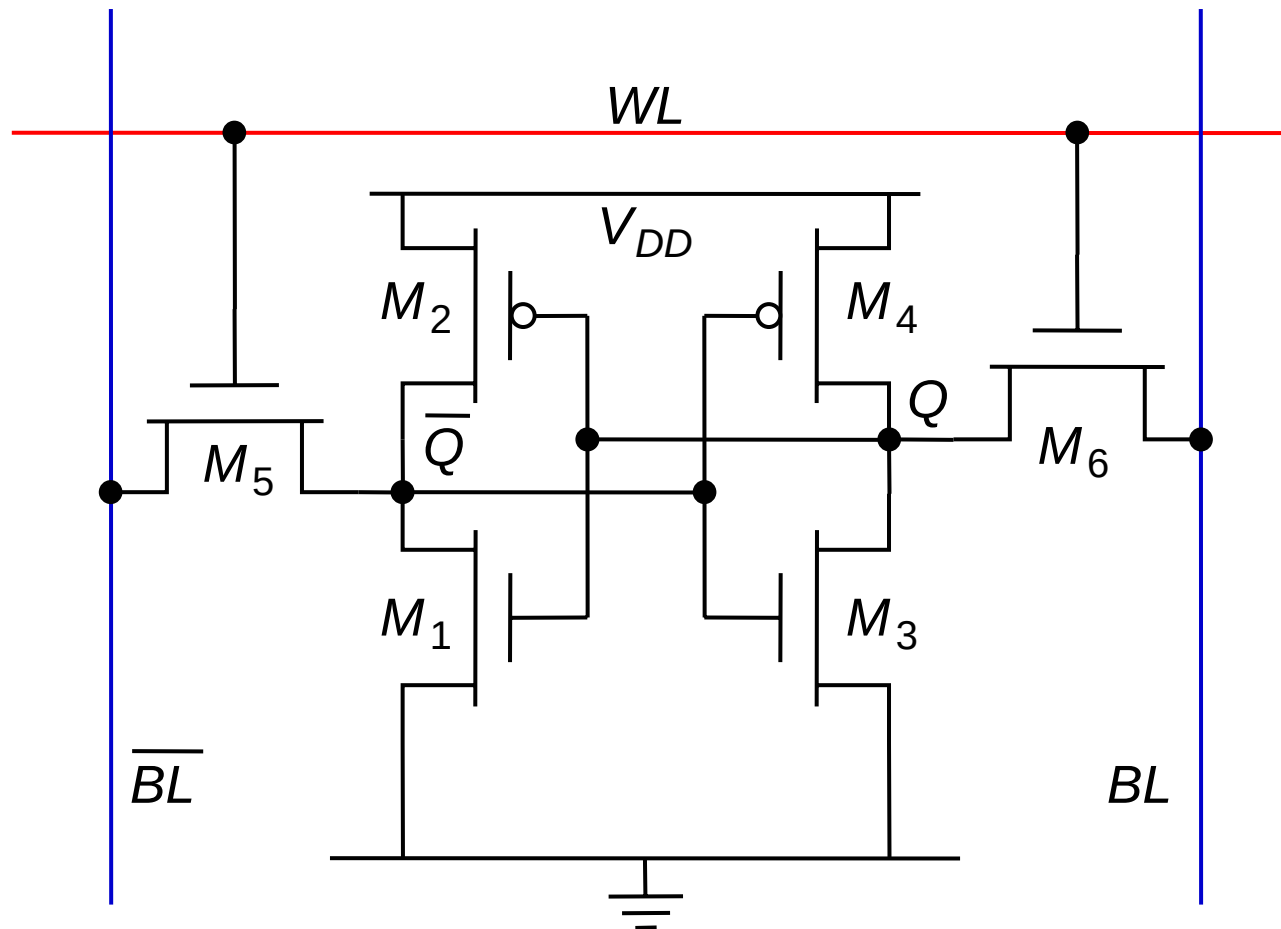
Periodic refresh required

Small (1-3 transistors/cell)

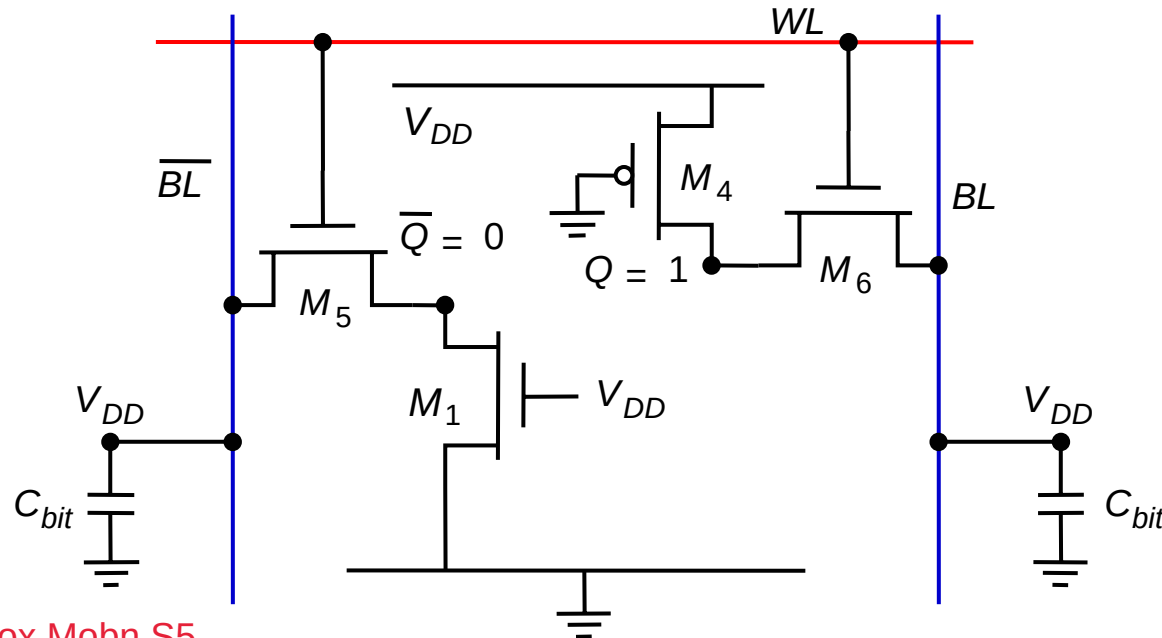
Slower

Single Ended

6-transistor CMOS SRAM Cell



CMOS SRAM Analysis (Read)



Kn,M5=Cox Mobn S5

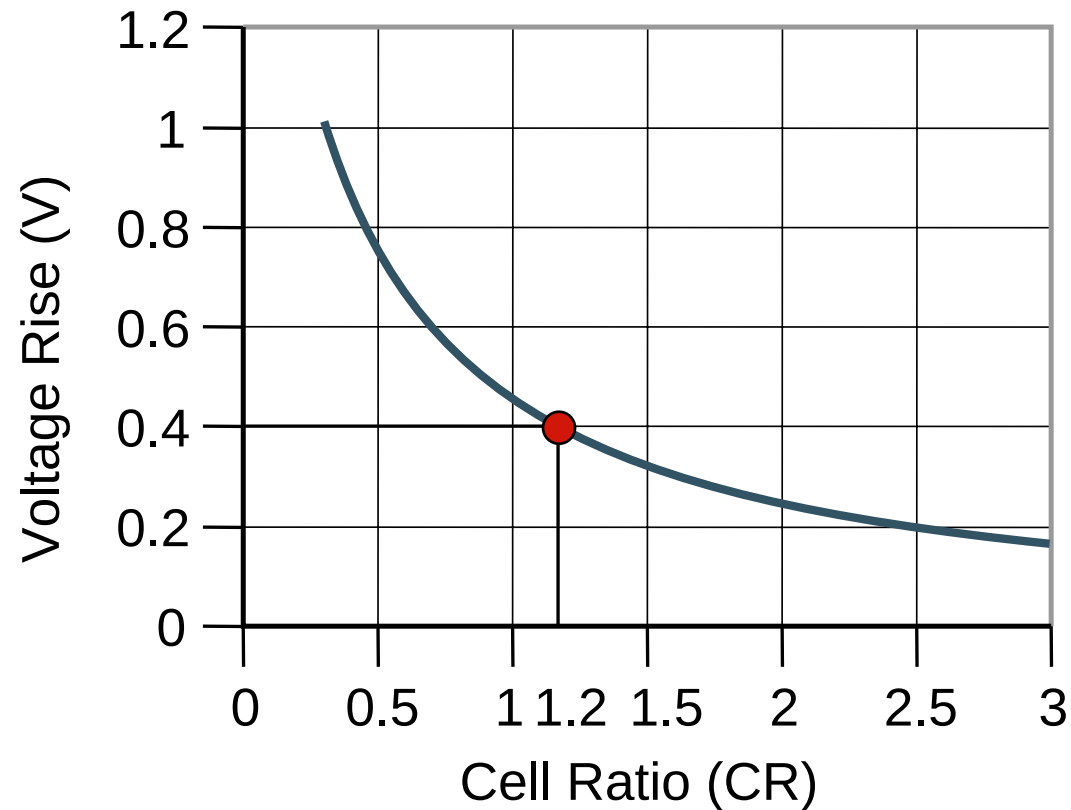
$$K_{n,M1} = C_{ox} \cdot M_{obn} \cdot S_1$$

$$k_{n,M5} \left((V_{DD} - \Delta V - V_{Tn}) V_{DSATn} - \frac{V_{DSATn}^2}{2} \right) \stackrel{S1}{=} k_{n,M1} \left((V_{DD} - V_{Tn}) \Delta V - \frac{\Delta V^2}{2} \right)$$

$$CR = S1/S5$$

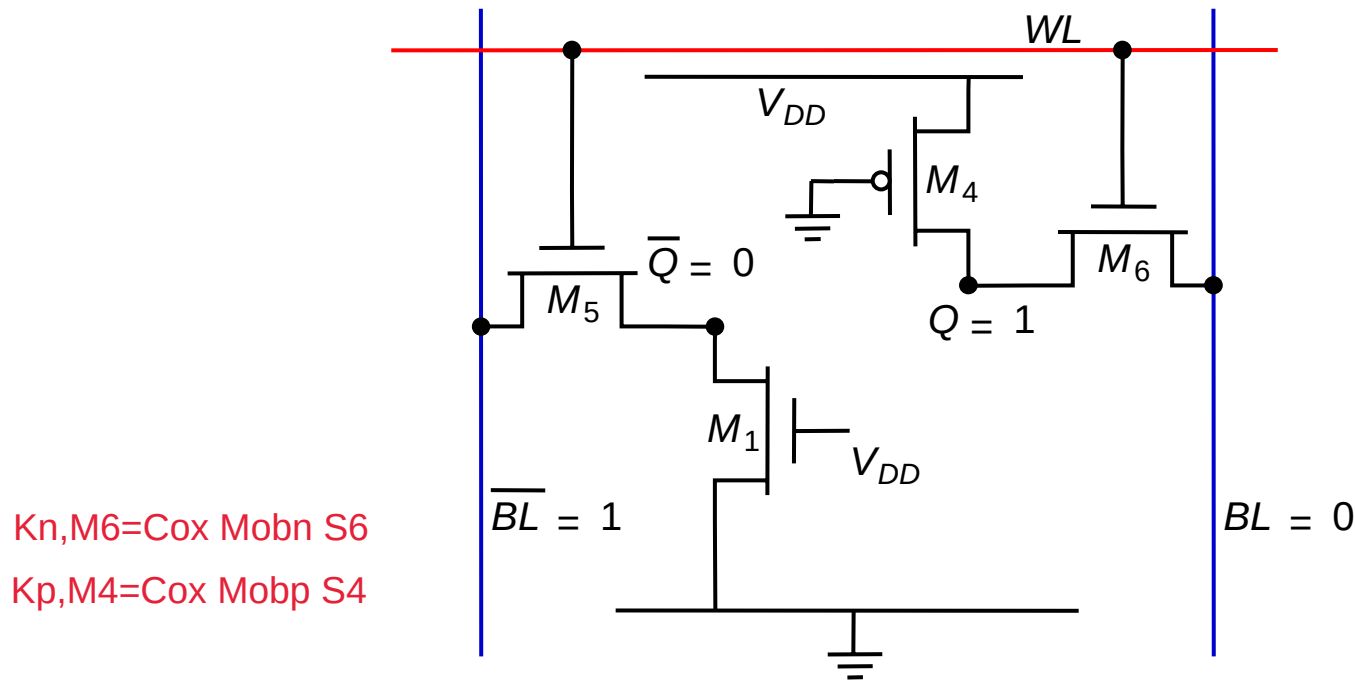
$$\Delta V = \frac{V_{DSATn} + CR(V_{DD} - V_{Tn}) - \sqrt{V_{DSATn}^2(1 + CR) + CR^2(V_{DD} - V_{Tn})^2}}{CR}$$

CMOS SRAM Analysis (Read)



$$CR = \frac{W_1/L_1}{W_5/L_5}$$

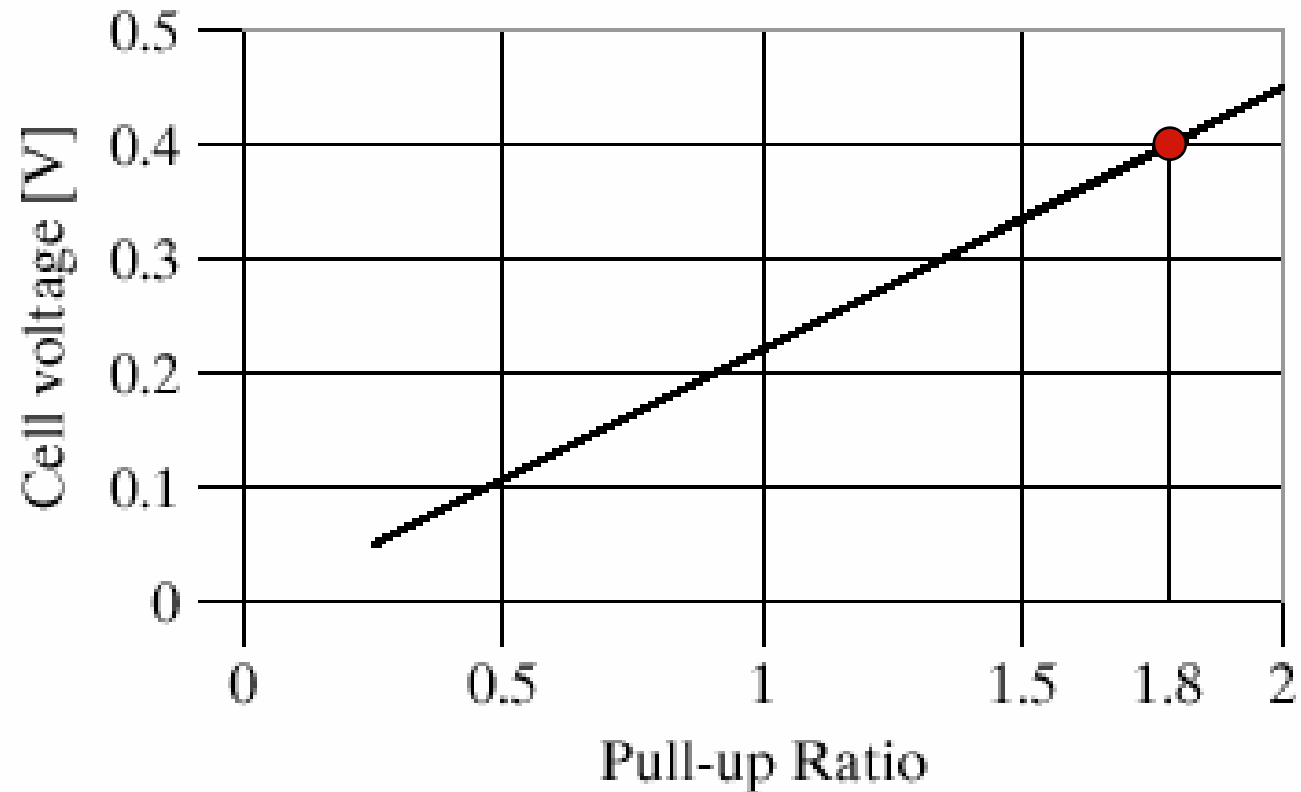
CMOS SRAM Analysis (Write)



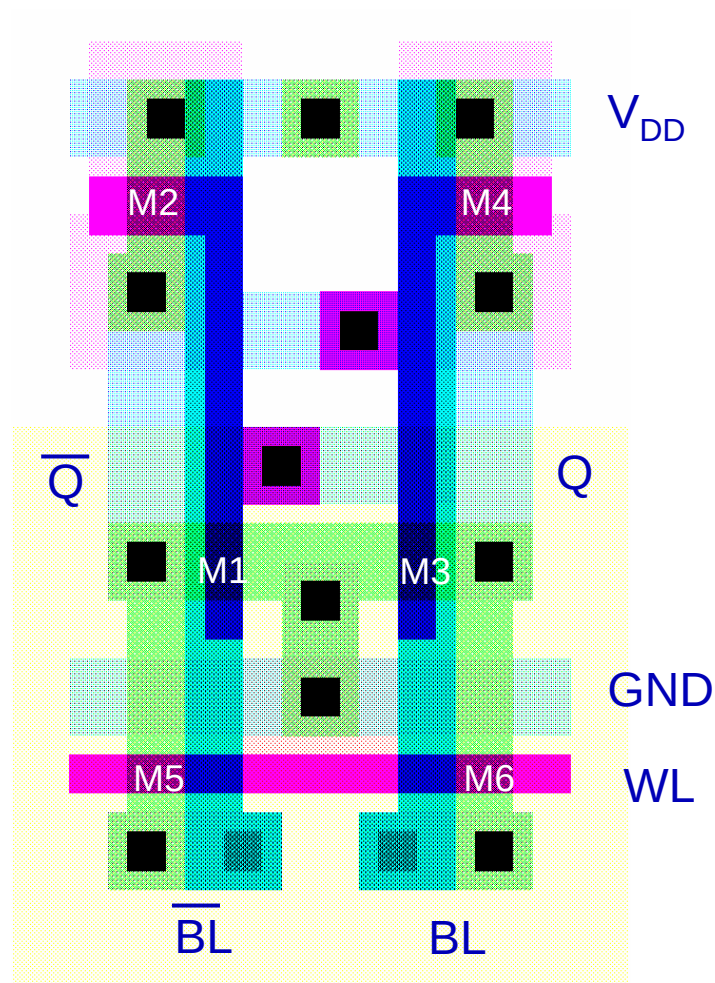
$$k_{n,M6} \left((V_{DD} - V_{Tn}) V_Q - \frac{V_Q^2}{2} \right) = k_{p,M4} \left((V_{DD} - |V_{Tp}|) V_{DSATp} - \frac{V_{DSATp}^2}{2} \right)$$

$$V_Q = V_{DD} - V_{Tn} - \sqrt{(V_{DD} - V_{Tn})^2 - 2 \frac{\mu_p}{\mu_n} PR \left((V_{DD} - |V_{Tp}|) V_{DSATp} - \frac{V_{DSATp}^2}{2} \right)}, \quad PR = S4/S6$$

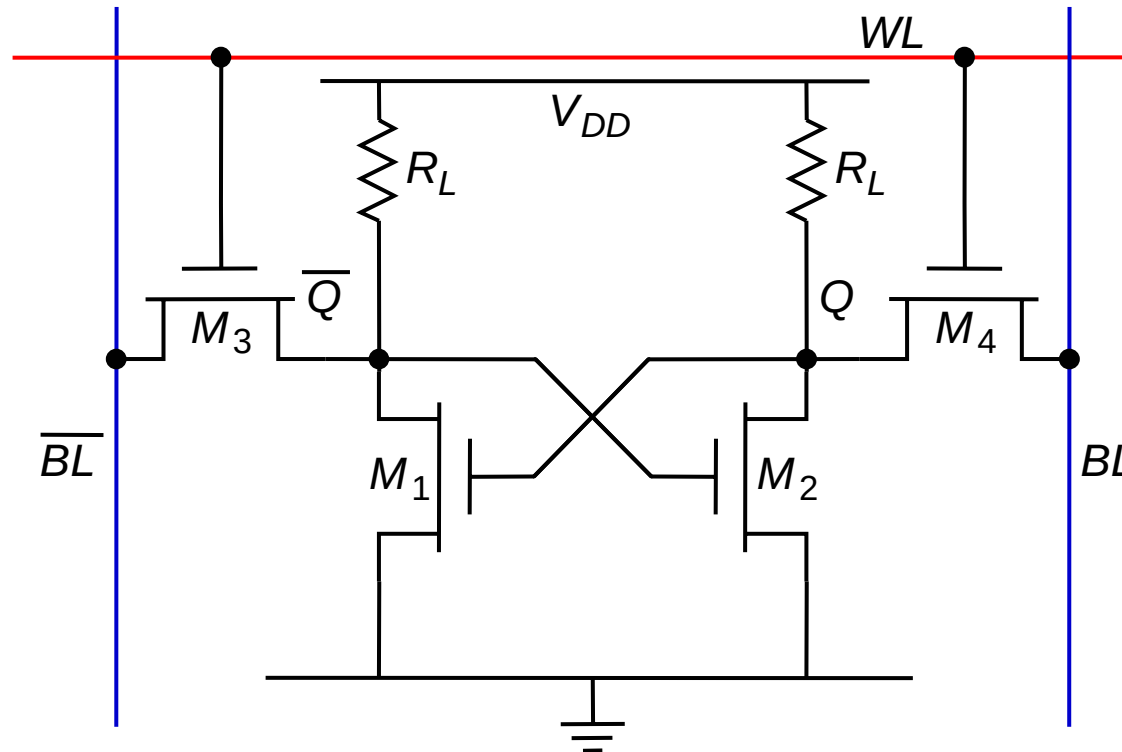
CMOS SRAM Analysis (Write)



6T-SRAM — Layout



Resistance-load SRAM Cell



Static power dissipation -- Want R_L large
Bit lines precharged to V_{DD} to address t_p problem

SRAM Characteristics

Table 12-2 Comparison of CMOS SRAM cells used in 1-Mbit memory
(from [Takada91])

	Complementary CMOS	Resistive Load	TFT Cell
Number of transistors	6	4	4 (+2 TFT)
Cell size	58.2 μm^2 (0.7- μm rule)	40.8 μm^2 (0.7- μm rule)	41.1 μm^2 (0.8- μm rule)
Standby current (per cell)	10^{-15} A	10^{-12} A	10^{-13} A

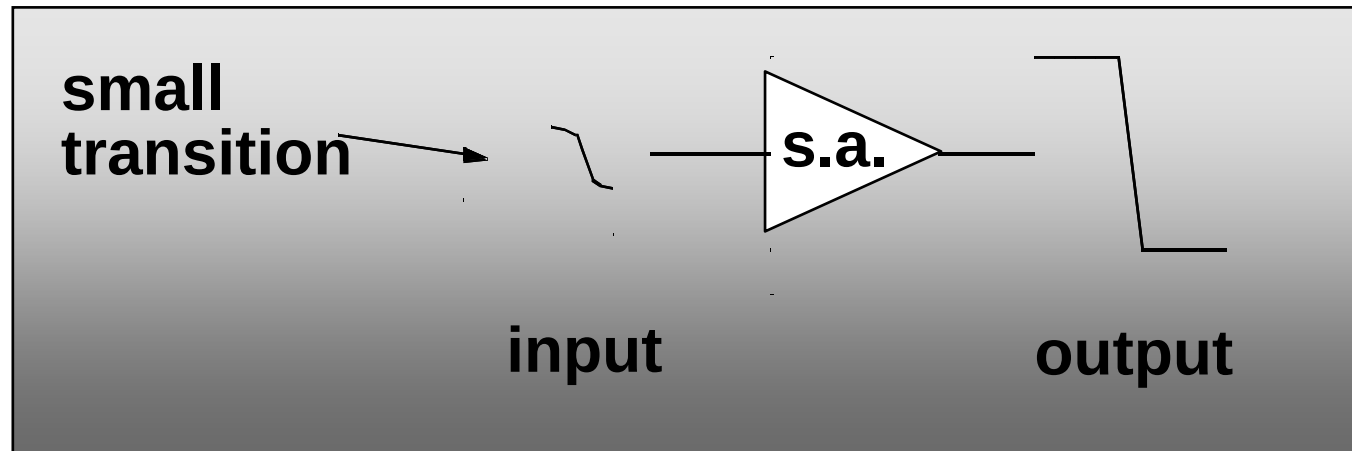
Sense Amplifiers

$$t_p = \frac{C \cdot \Delta V}{I_{av}}$$

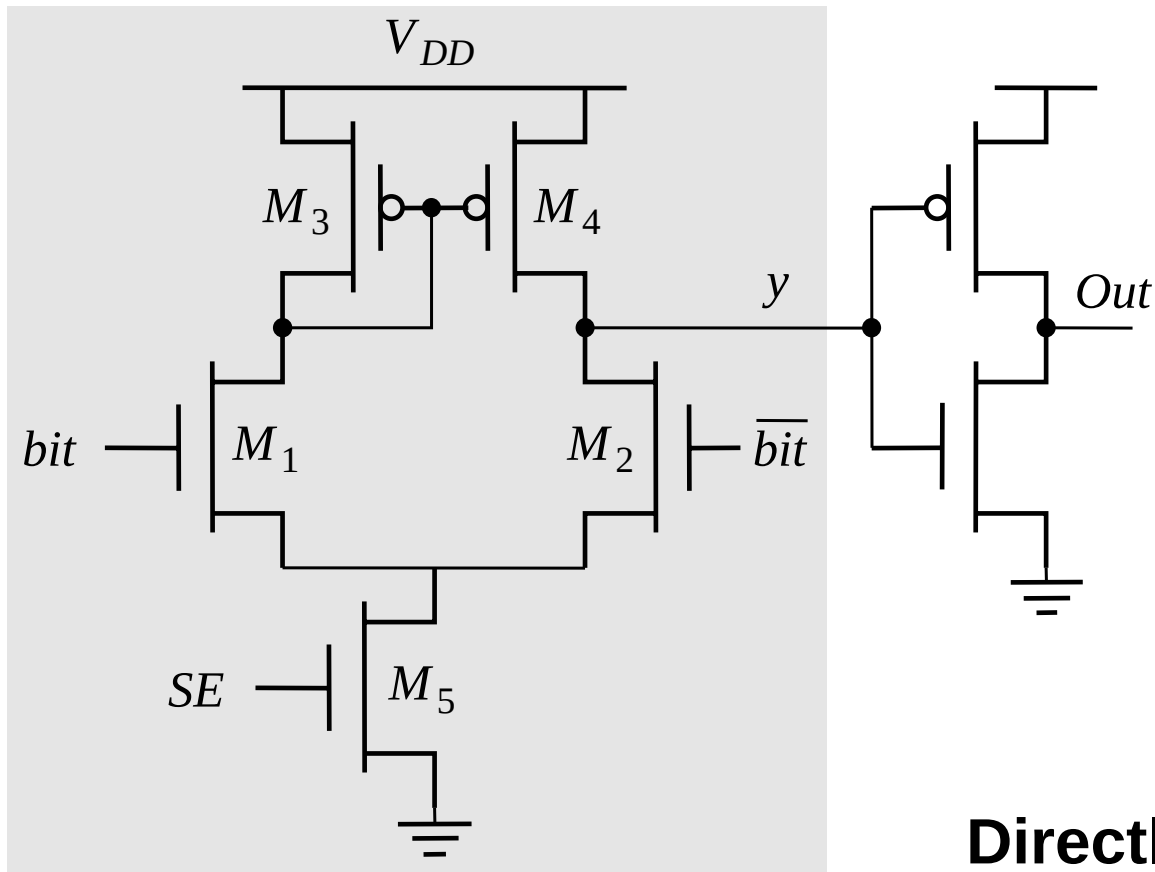
Annotations:

- make ΔV as small as possible (points to ΔV)
- large (points to C)
- small (points to I_{av})

Idea: Use Sense Amplifier

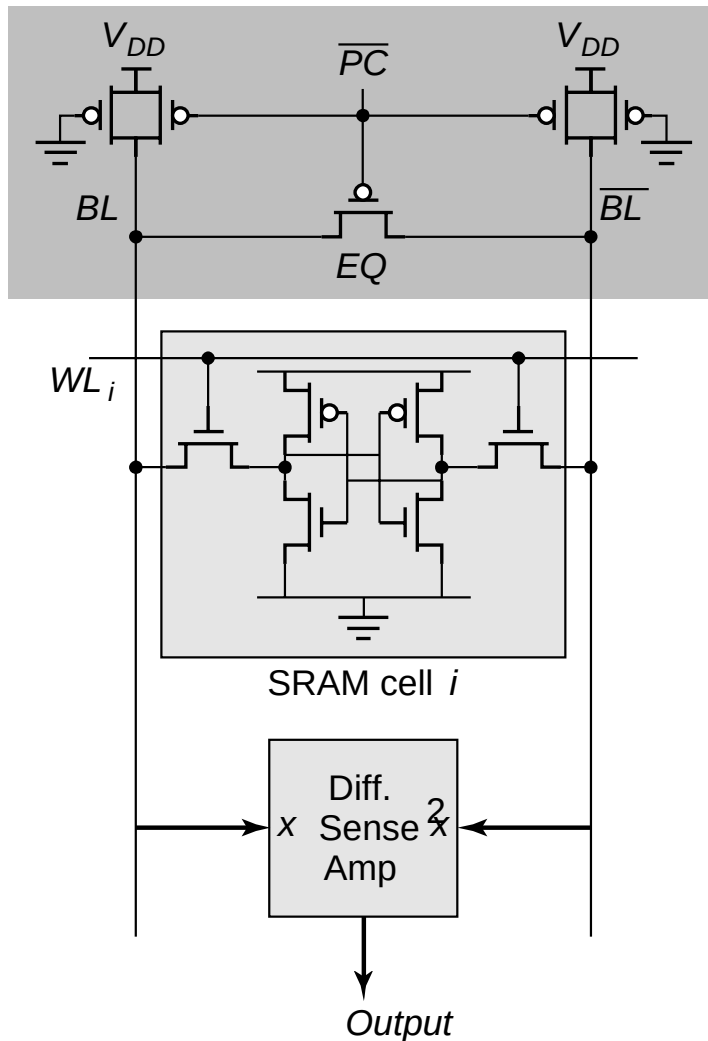


Differential Sense Amplifier

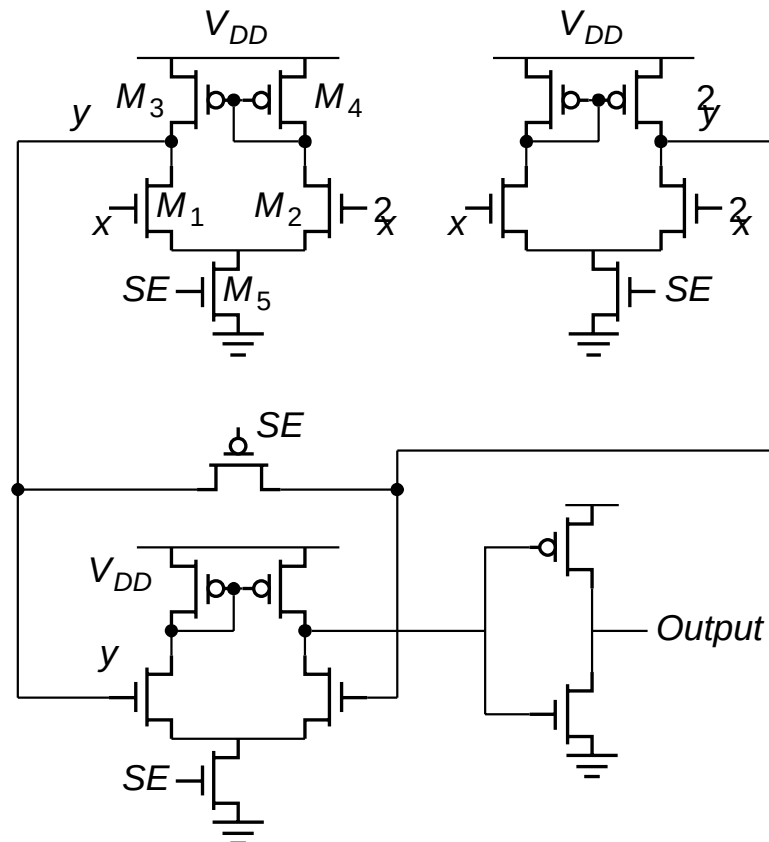


**Directly applicable to
SRAMs**

Differential Sensing – SRAM

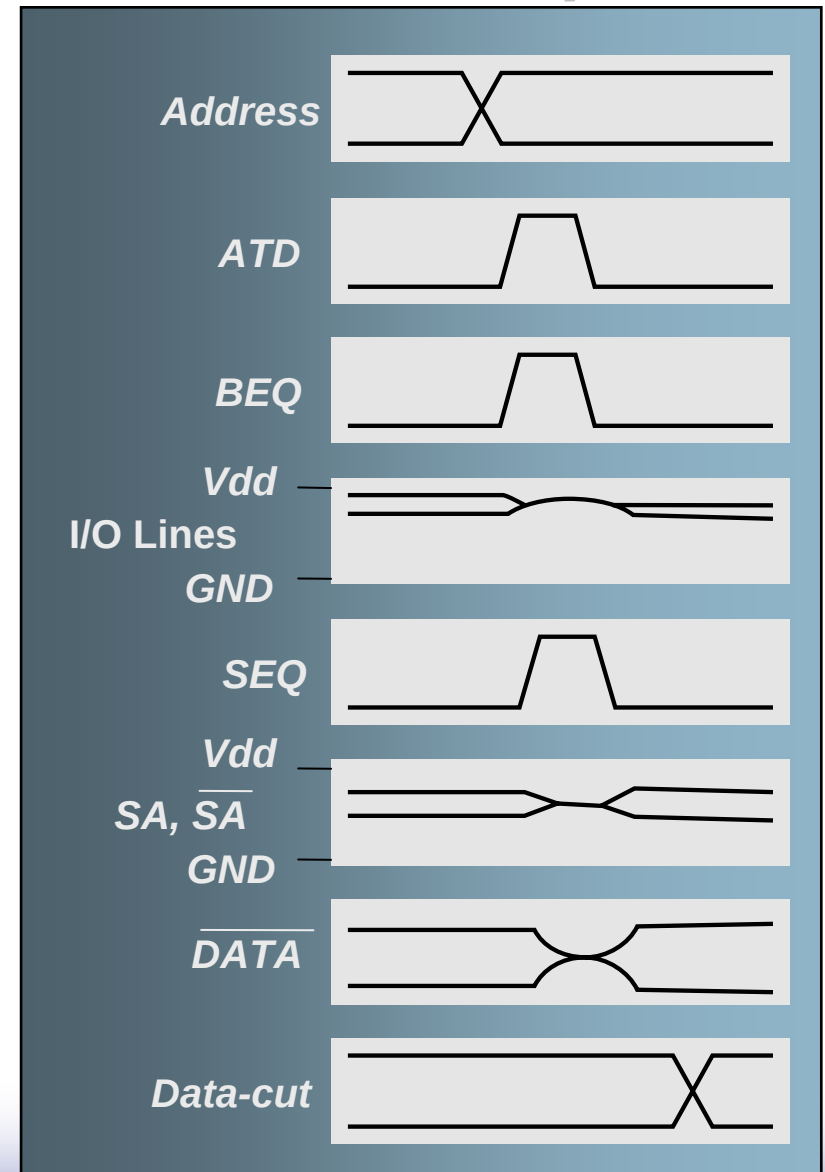
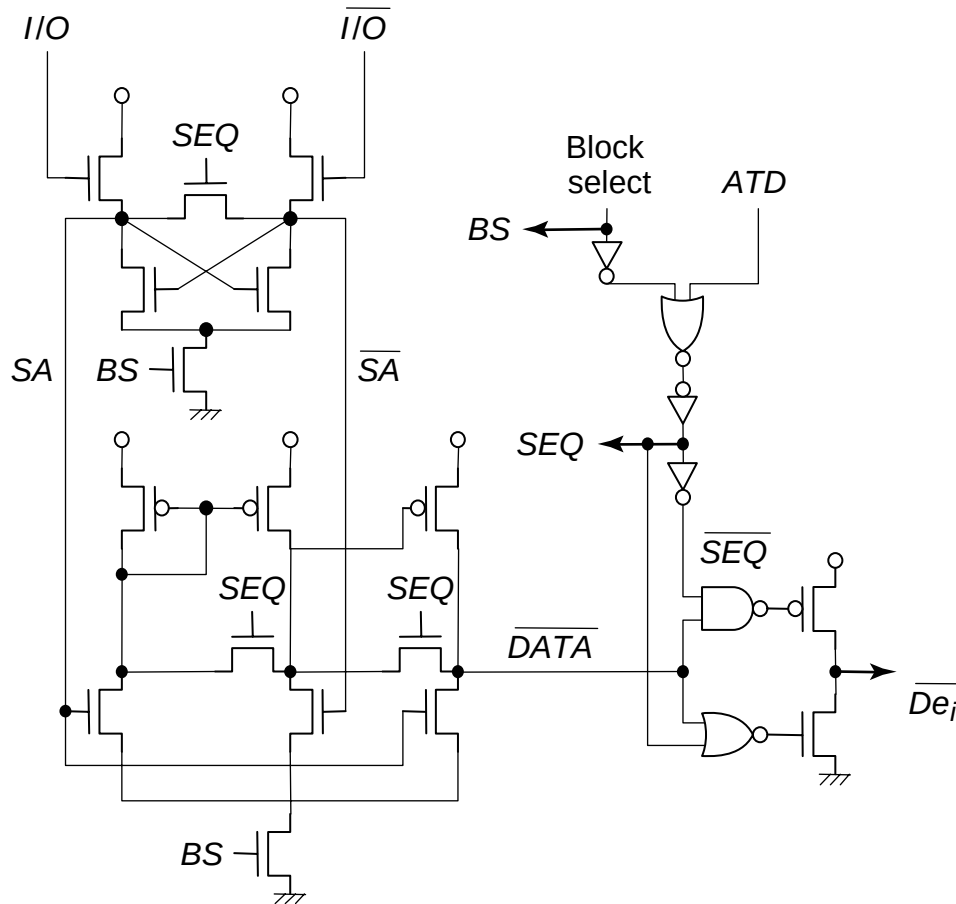


(a) SRAM sensing scheme

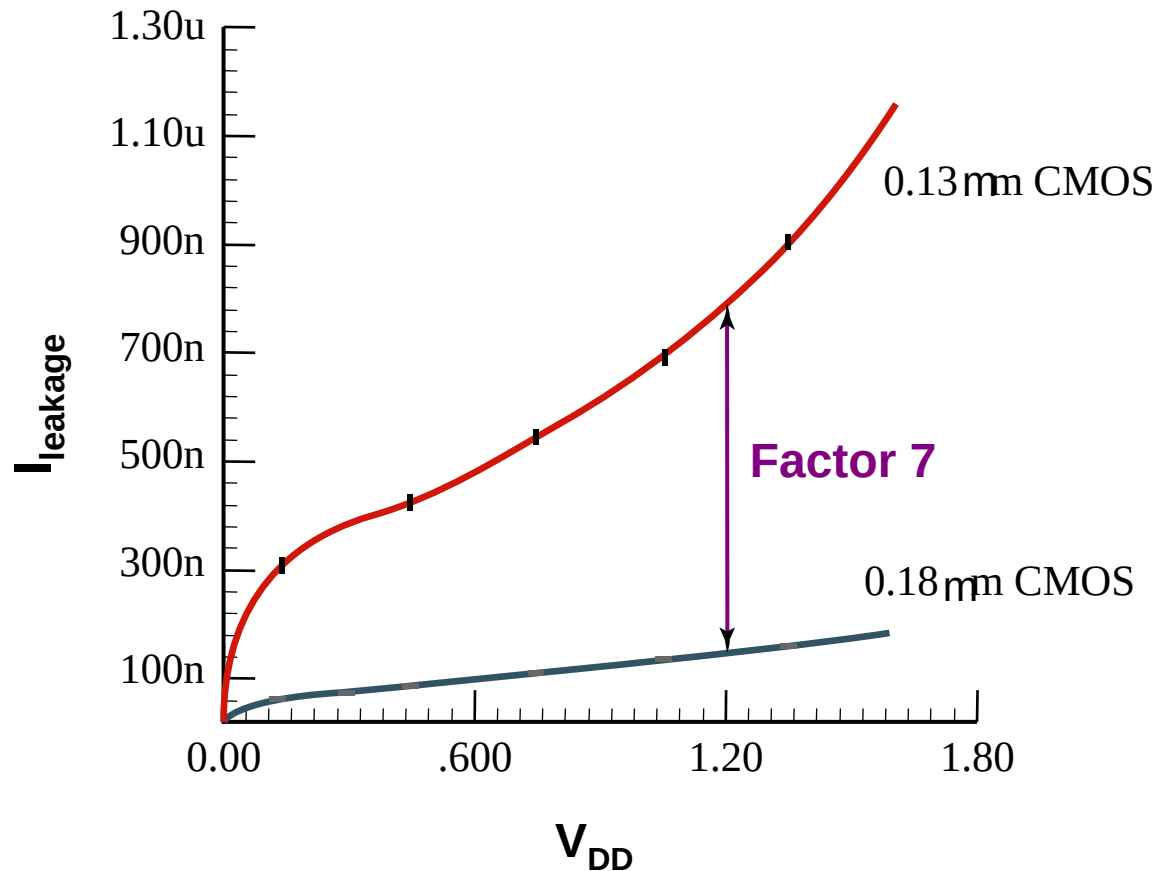


(b) two stage differential amplifier

Sense Amplifier (and Waveforms)

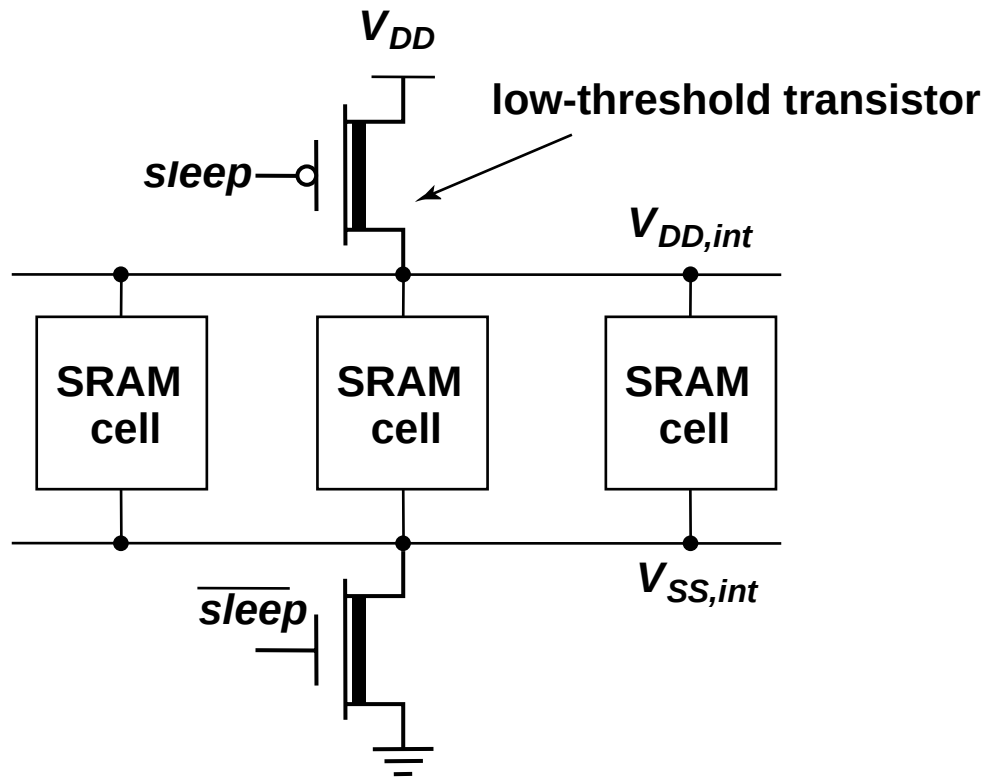


Data Retention in SRAM

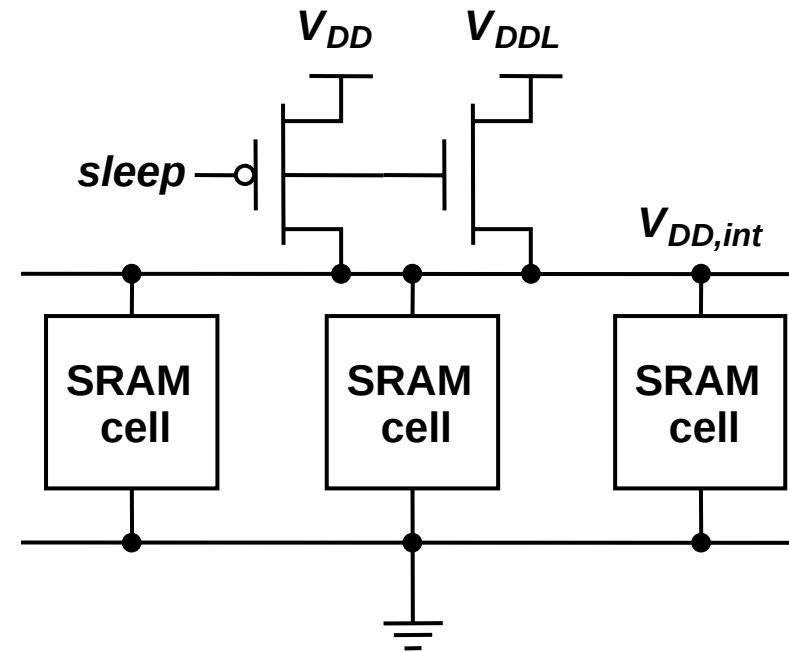


SRAM leakage increases with technology scaling

Suppressing Leakage in SRAM

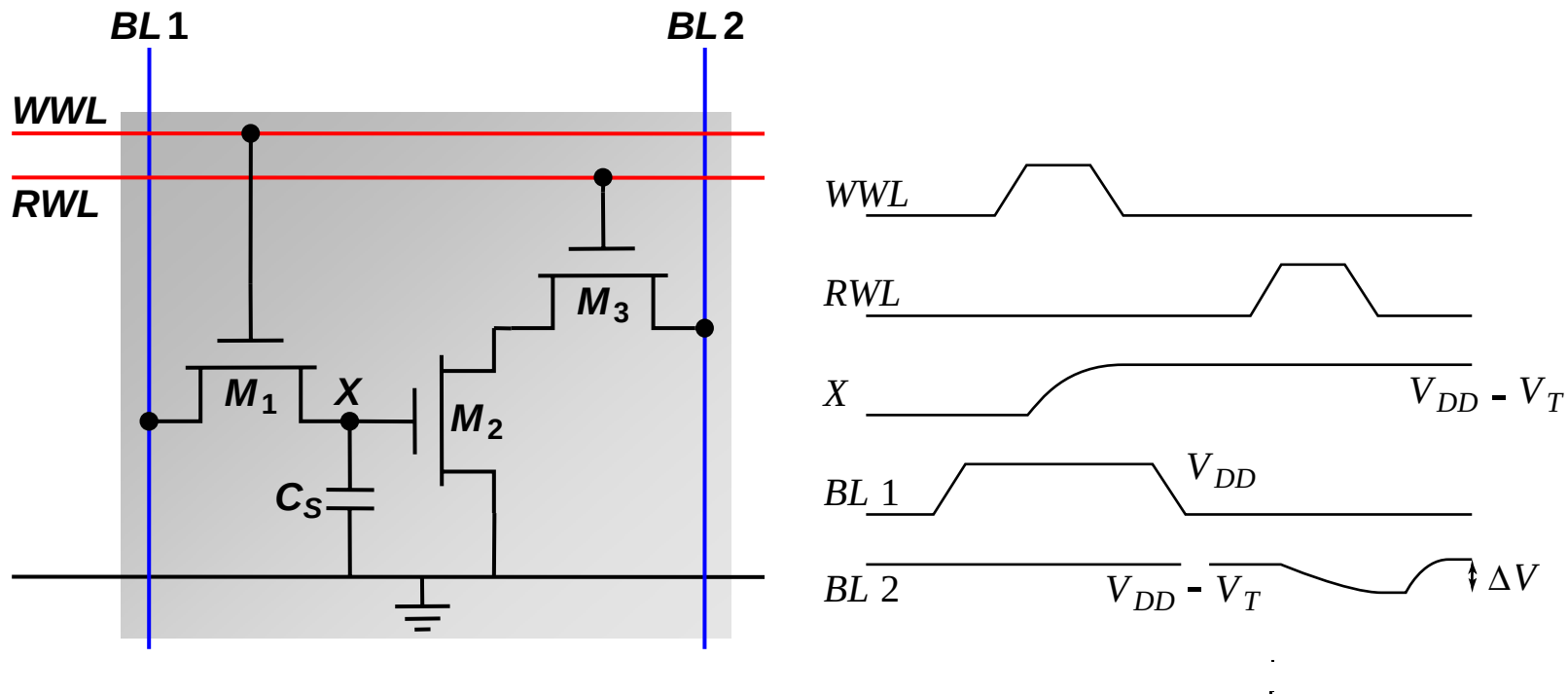


Inserting Extra Resistance



Reducing the supply voltage

3-Transistor DRAM Cell

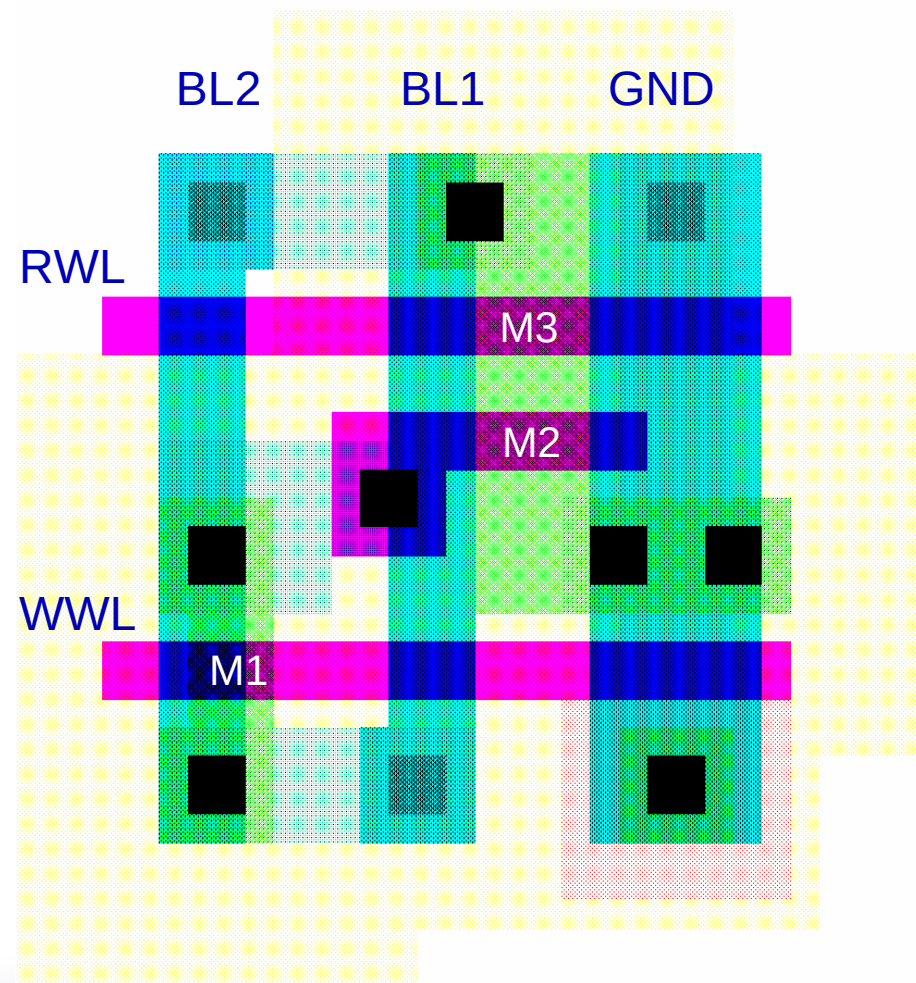


No constraints on device ratios

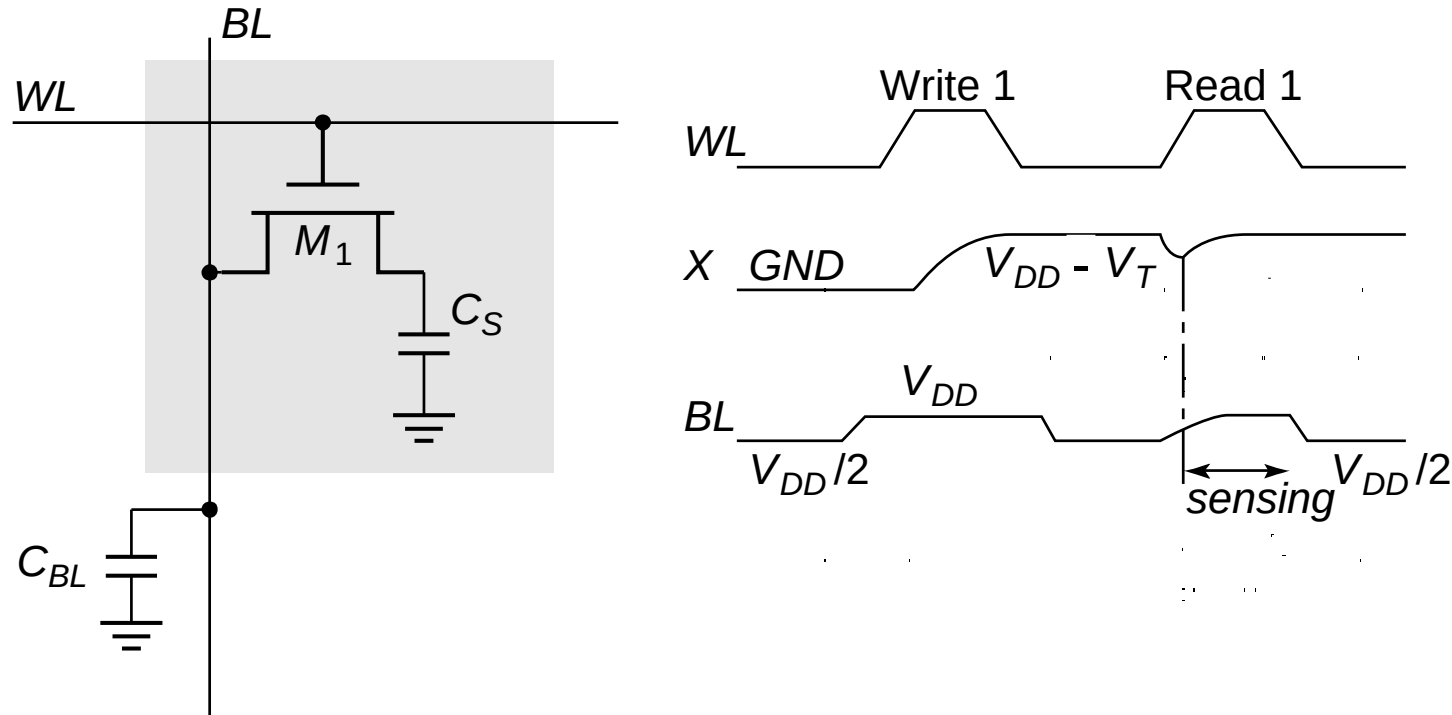
Reads are non-destructive

Value stored at node X when writing a "1" = $V_{WWL} - V_{Tn}$

3T-DRAM — Layout



1-Transistor DRAM Cell



Write: C_S is charged or discharged by asserting WL and BL.

Read: Charge redistribution takes places between bit line and storage capacitance

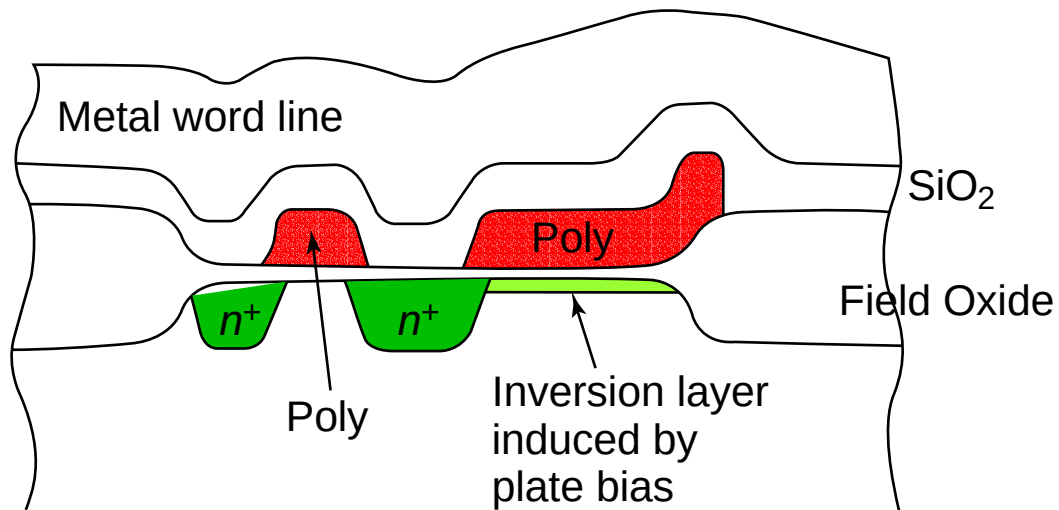
$$\Delta V = V_{BL} - V_{PRE} = V_{BIT} - V_{PRE} \frac{C_S}{C_S + C_{BL}}$$

Voltage swing is small; typically around 250 mV.

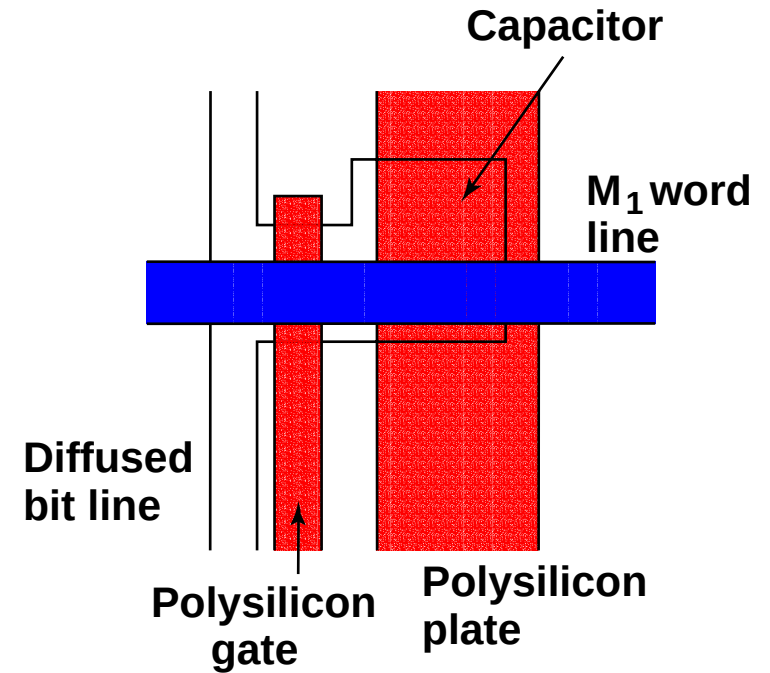
DRAM Cell Observations

- ❑ 1T DRAM requires a sense amplifier for each bit line, due to charge redistribution read-out.
- ❑ DRAM memory cells are single ended in contrast to SRAM cells.
- ❑ The read-out of the 1T DRAM cell is destructive; read and refresh operations are necessary for correct operation.
- ❑ Unlike 3T cell, 1T cell requires presence of an extra capacitance that must be explicitly included in the design.
- ❑ When writing a “1” into a DRAM cell, a threshold voltage is lost. This charge loss can be circumvented by bootstrapping the word lines to a higher value than V_{DD}

1-T DRAM Cell



Cross-section



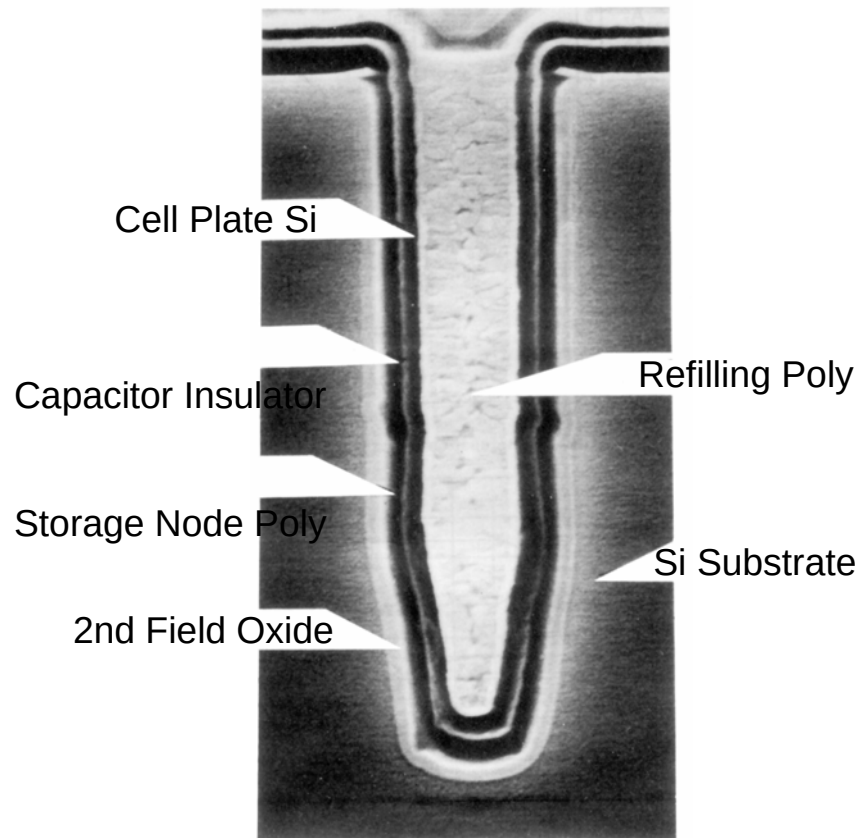
Layout

Uses Polysilicon-Diffusion Capacitance
Expensive in Area

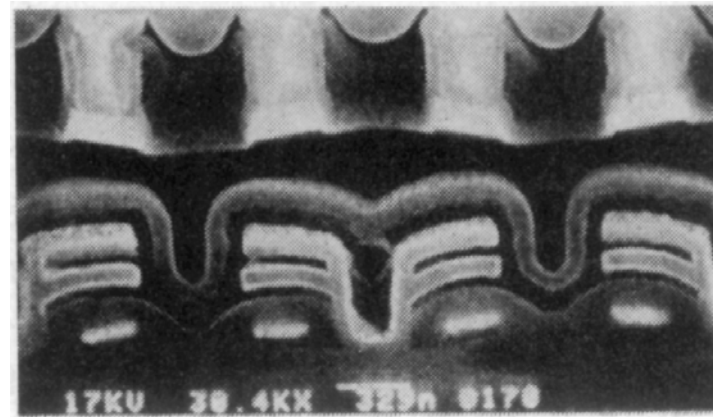
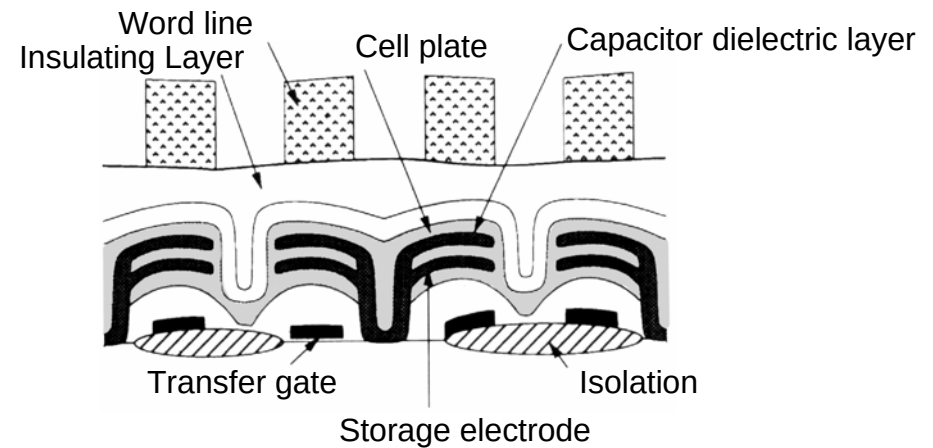
SEM of poly-diffusion capacitor 1T-DRAM



Advanced 1T DRAM Cells

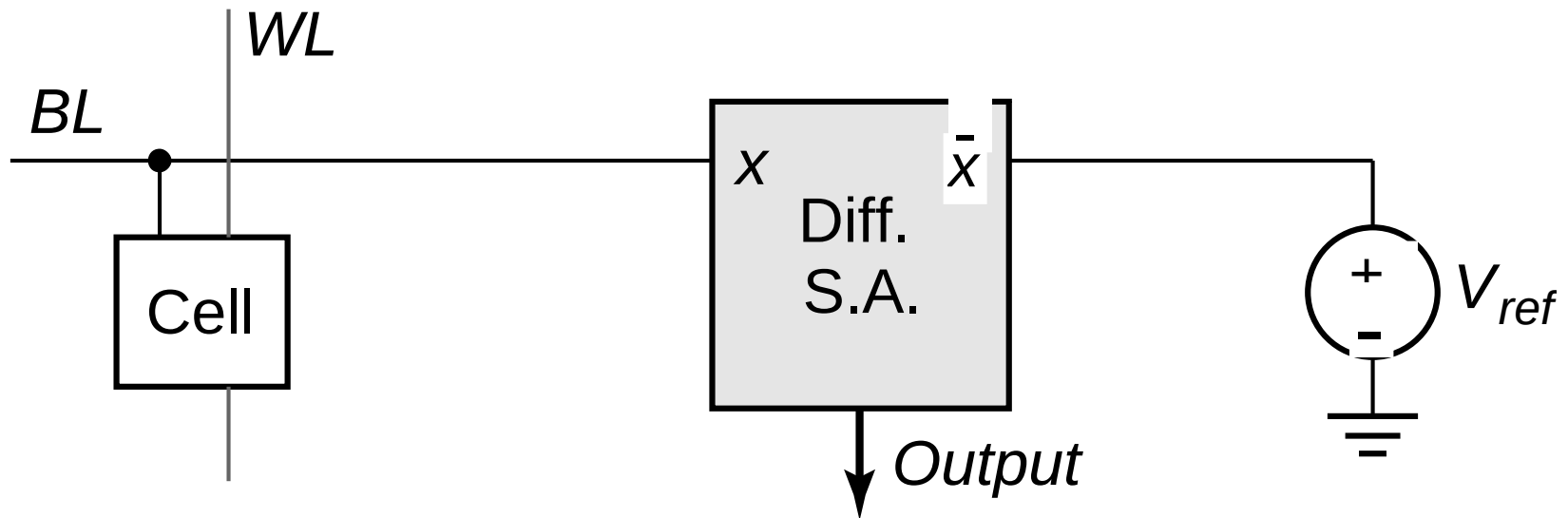


Trench Cell



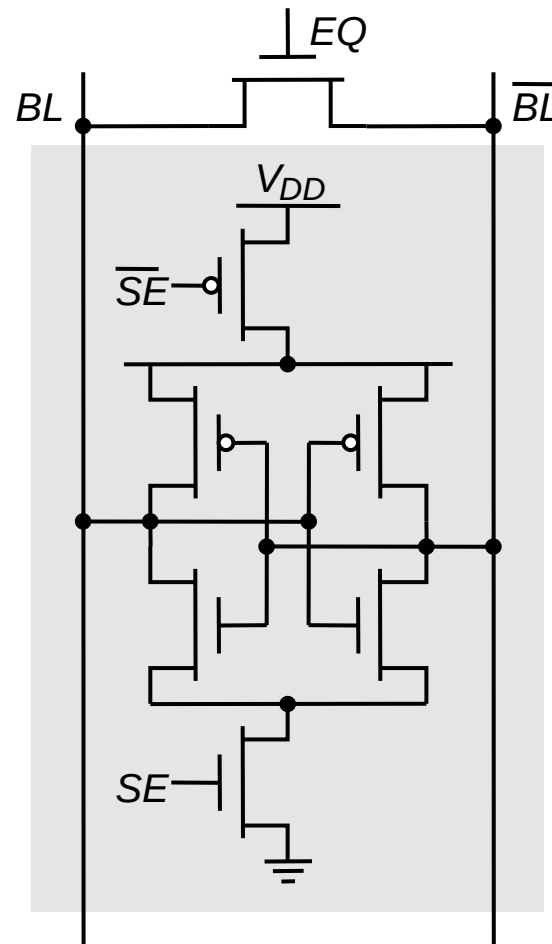
Stacked-capacitor Cell

Single-to-Differential Conversion



How to make a good V_{ref} ?

Latch-Based Sense Amplifier (DRAM)

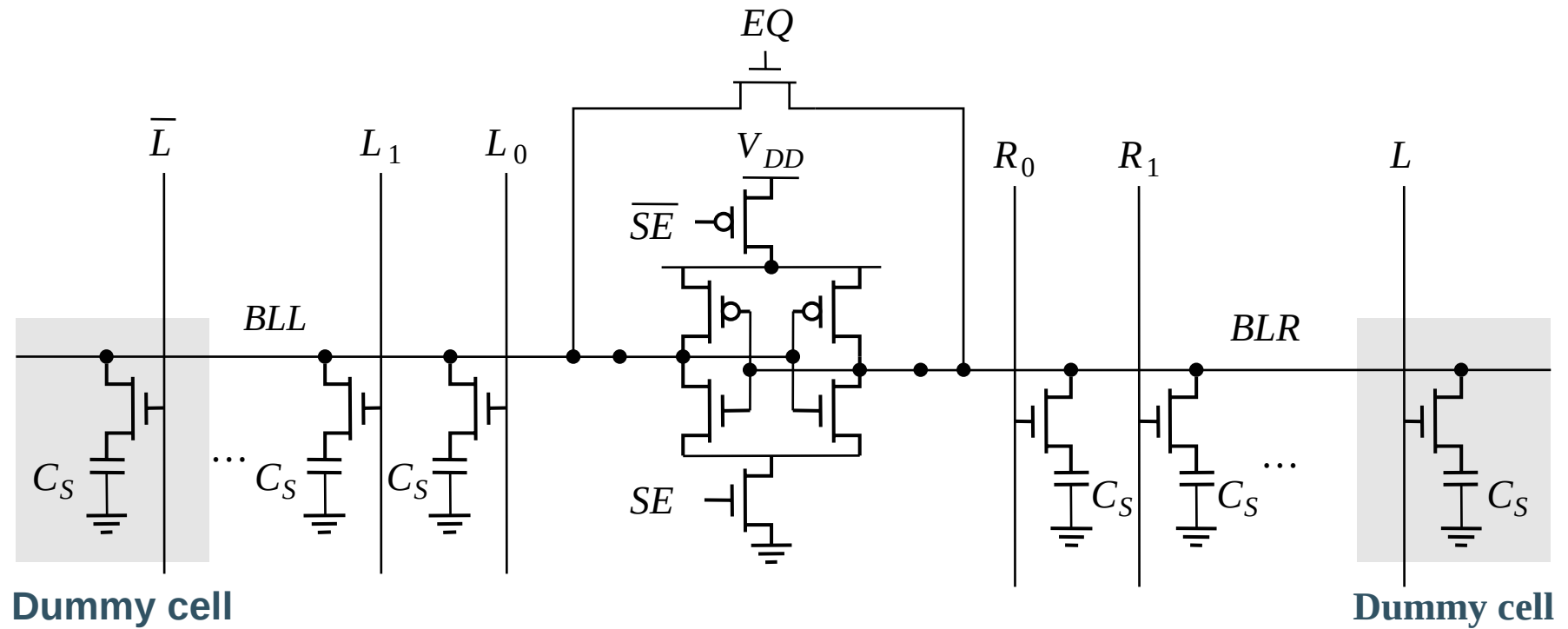


Initialized in its meta-stable point with EQ

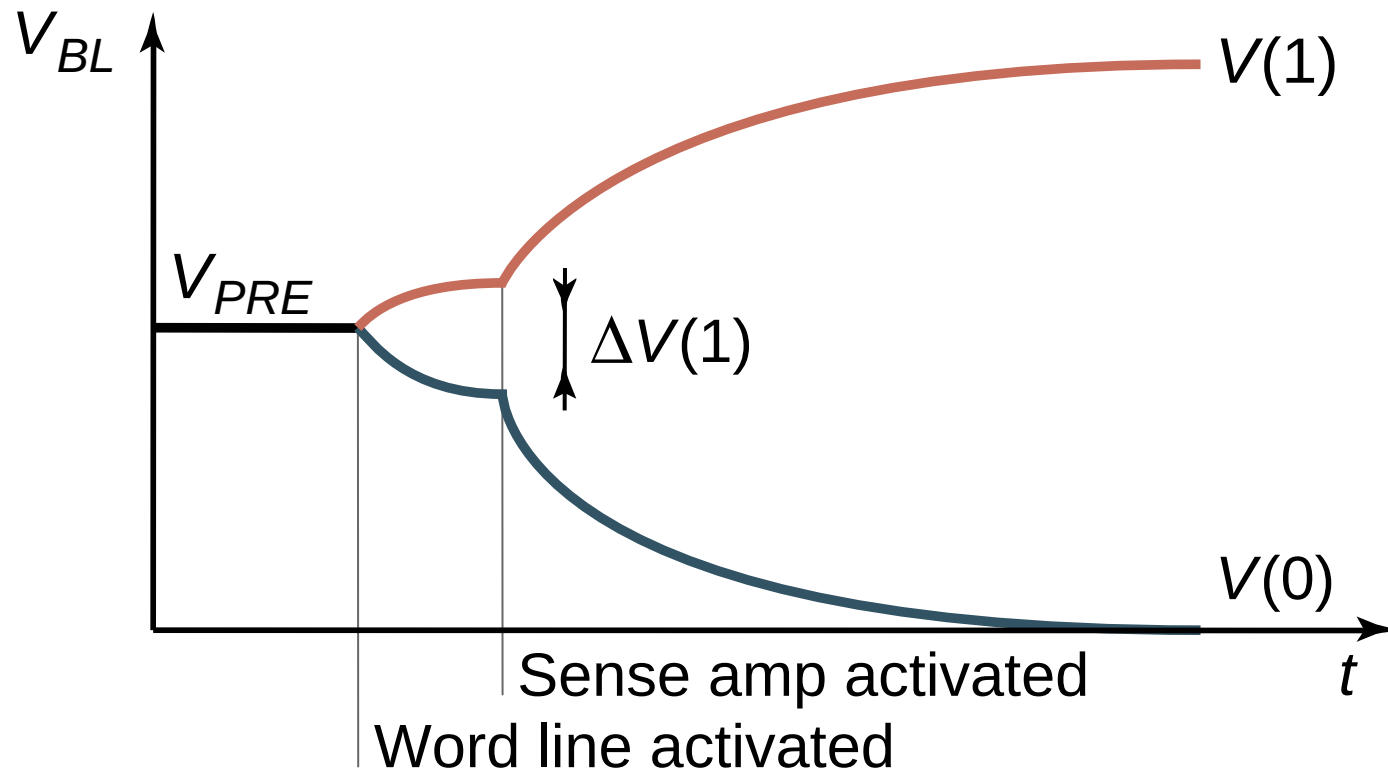
Once adequate voltage gap created, sense amp enabled with SE

Positive feedback quickly forces output to a stable operating point.

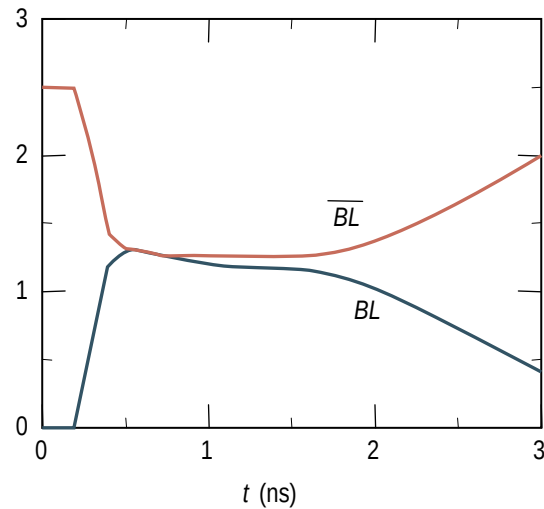
Open bitline architecture with dummy cells



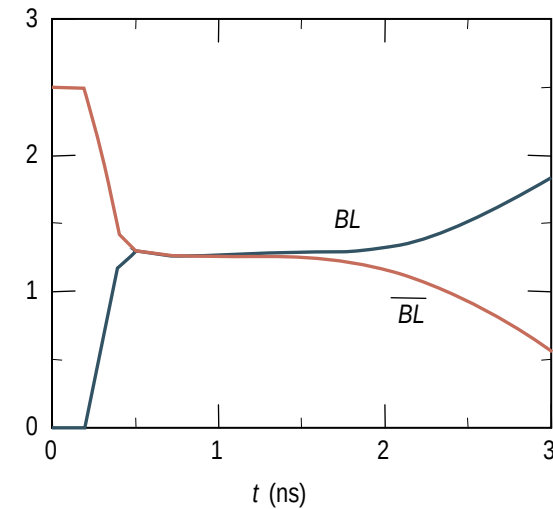
Sense Amp Operation



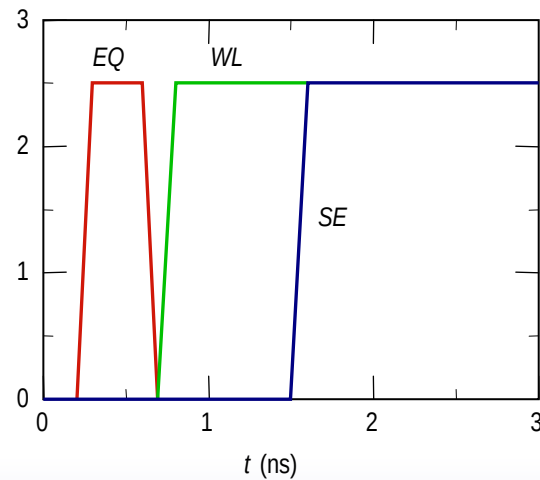
DRAM Read Process with Dummy Cell



reading 0



reading 1



control signals

Periphery

- ☐ **Decoders**
- ☐ **Sense Amplifiers**
- ☐ **Input/Output Buffers**
- ☐ **Control / Timing Circuitry**

Row Decoders

Collection of 2^M complex logic gates
Organized in regular and dense fashion

(N)AND Decoder

$$WL_0 = A_0 A_1 A_2 A_3 A_4 A_5 A_6 A_7 A_8 A_9$$

$$WL_{511} = \bar{A}_0 A_1 A_2 A_3 A_4 A_5 A_6 A_7 A_8 A_9$$

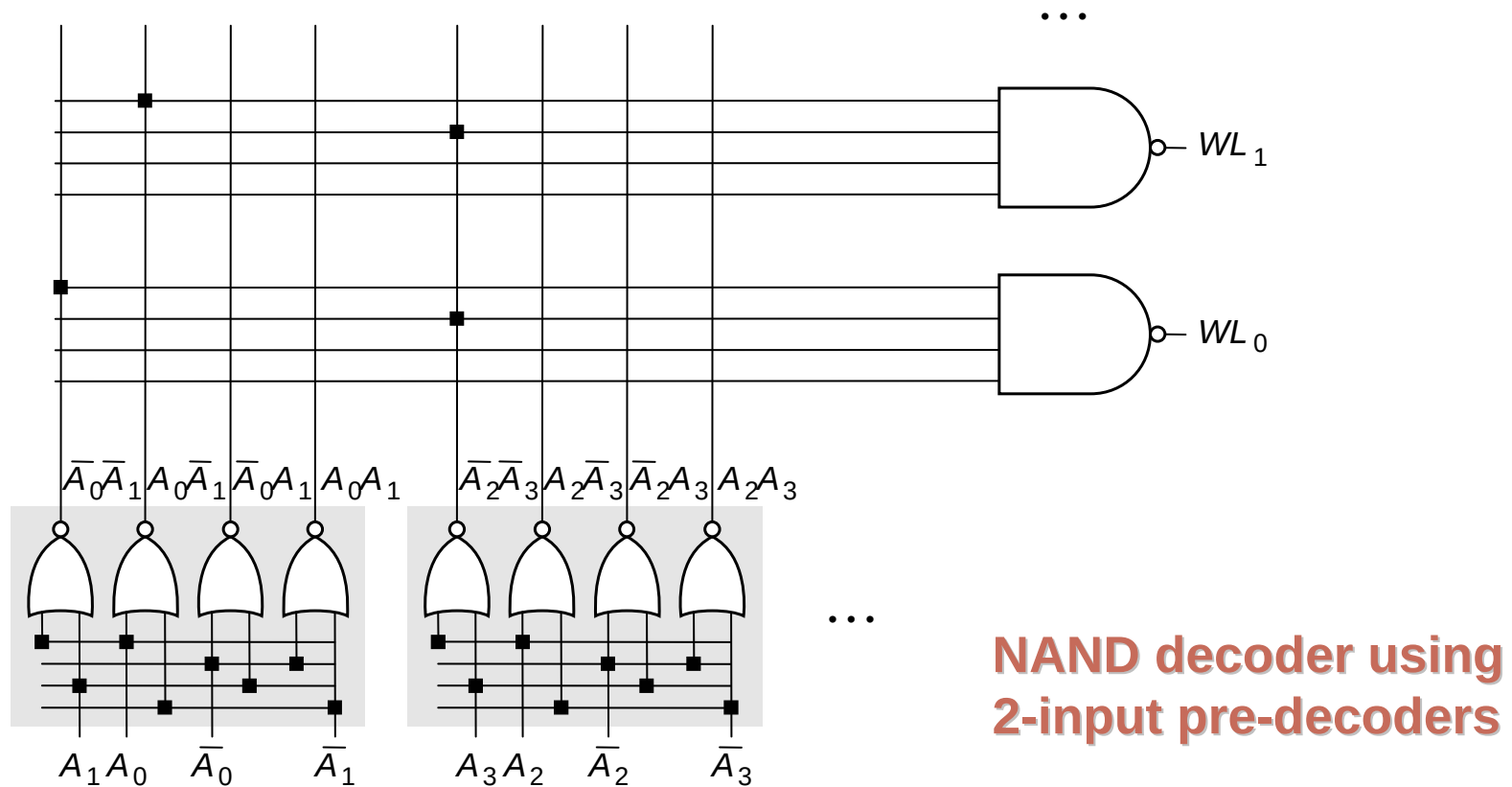
NOR Decoder

$$WL_0 = \overline{A_0 + A_1 + A_2 + A_3 + A_4 + A_5 + A_6 + A_7 + A_8 + A_9}$$

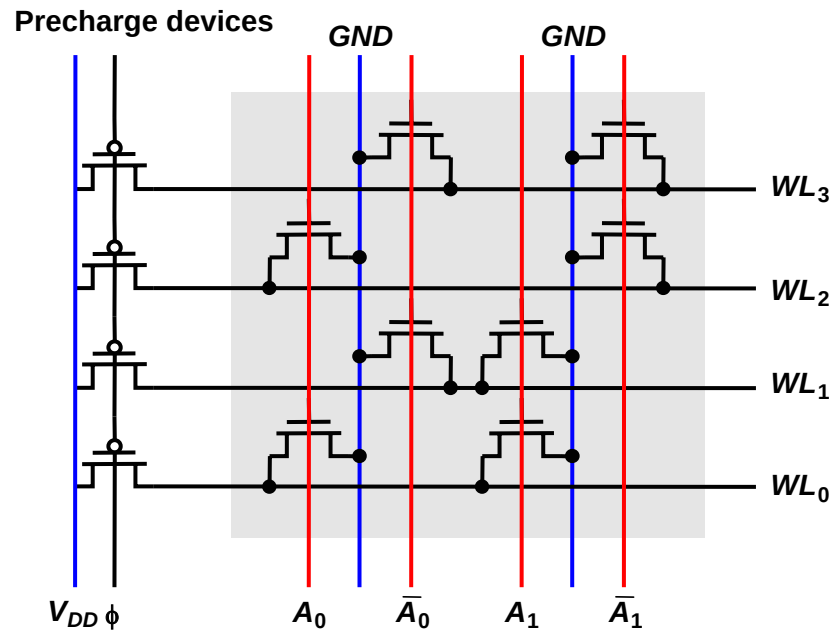
$$WL_{511} = \overline{A_0 + \bar{A}_1 + \bar{A}_2 + \bar{A}_3 + \bar{A}_4 + \bar{A}_5 + \bar{A}_6 + \bar{A}_7 + \bar{A}_8 + \bar{A}_9}$$

Hierarchical Decoders

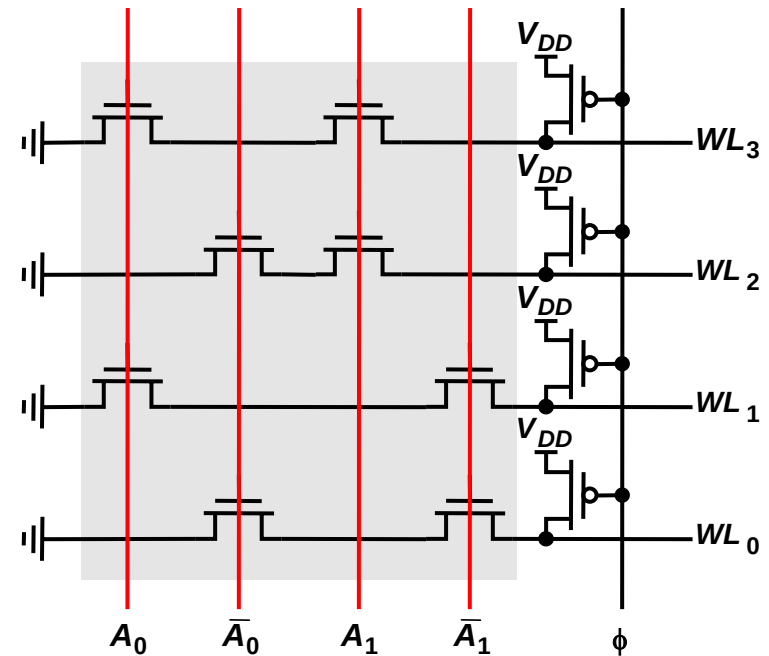
Multi-stage implementation improves performance



Dynamic Decoders

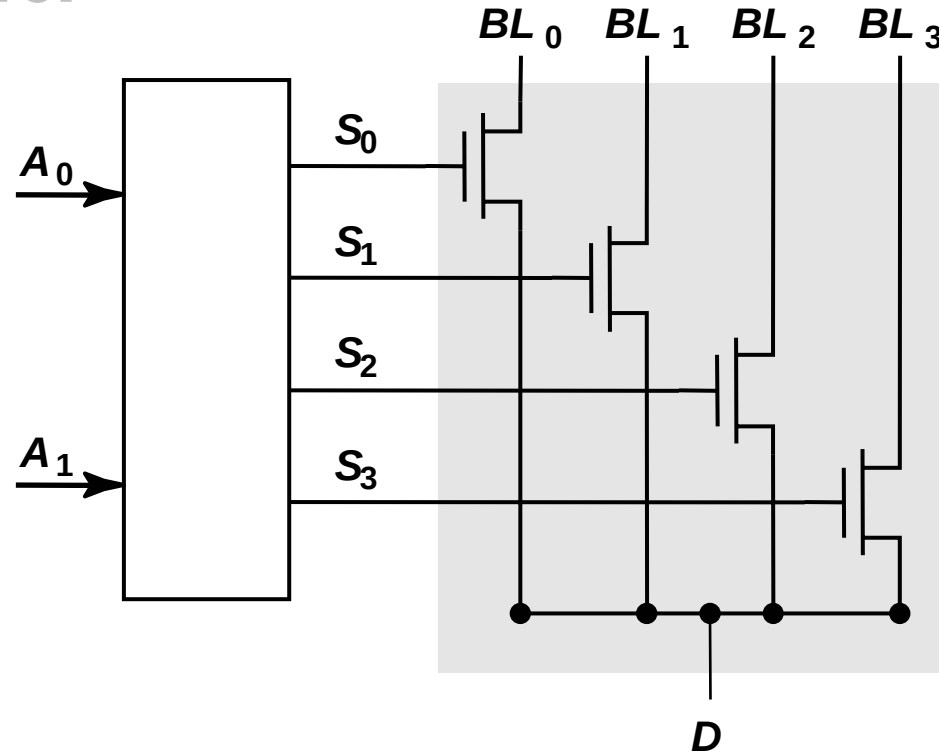


2-input NOR decoder



2-input NAND decoder

4-input pass-transistor based column decoder

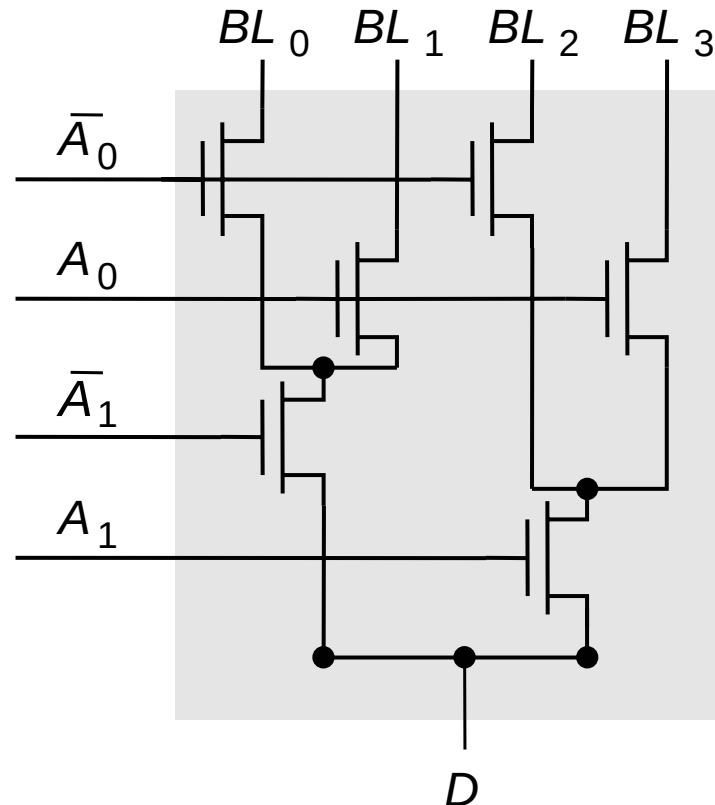


Advantages: speed (t_{pd} does not add to overall memory access time)

Only one extra transistor in signal path

Disadvantage: Large transistor count

4-to-1 tree based column decoder



Number of devices drastically reduced

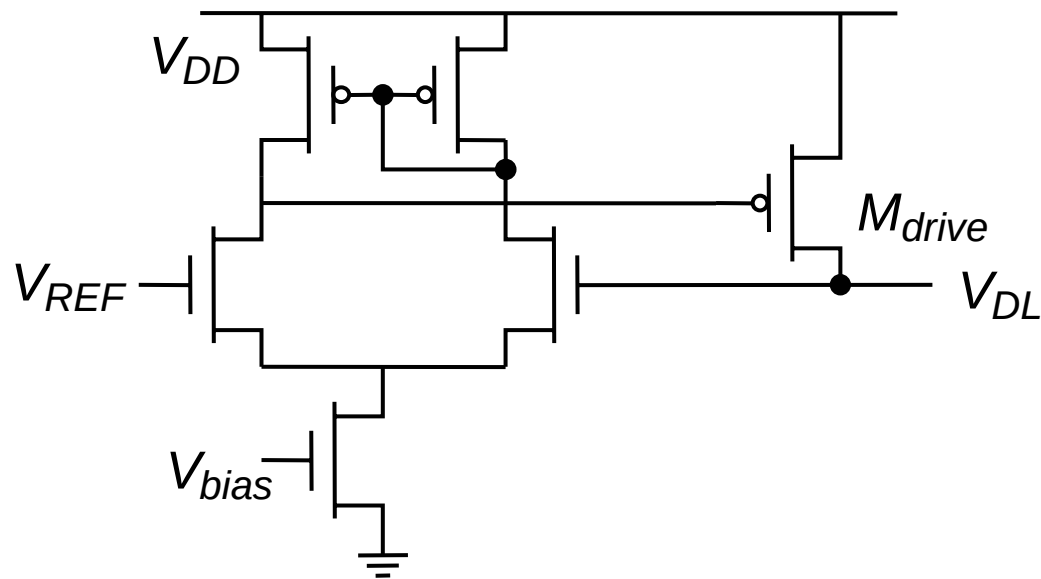
Delay increases quadratically with # of sections; prohibitive for large decoders

Solutions: buffers

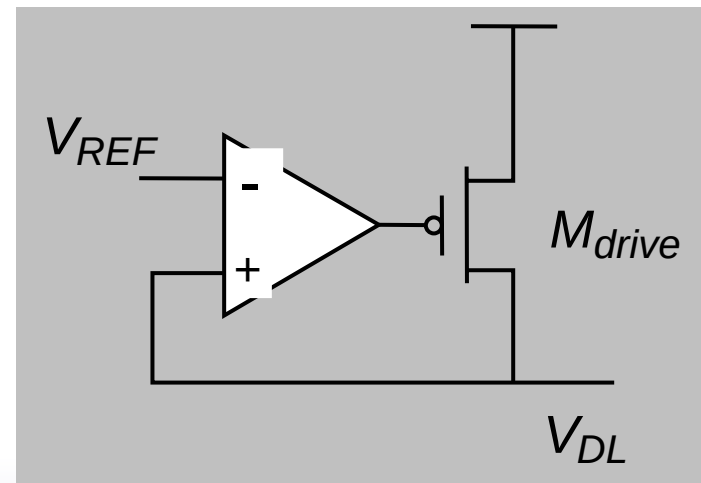
progressive sizing

combination of tree and pass transistor approaches

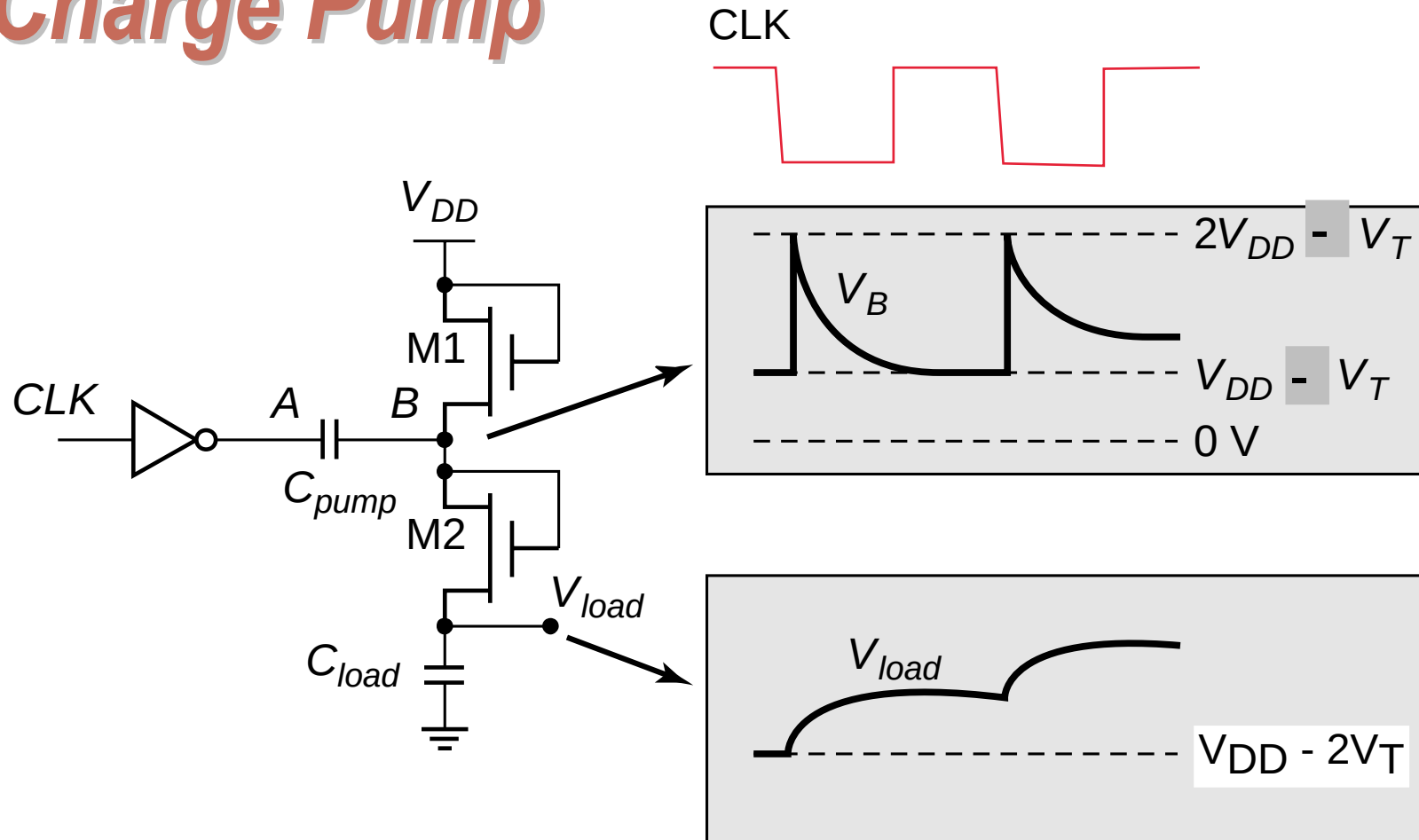
Voltage Regulator



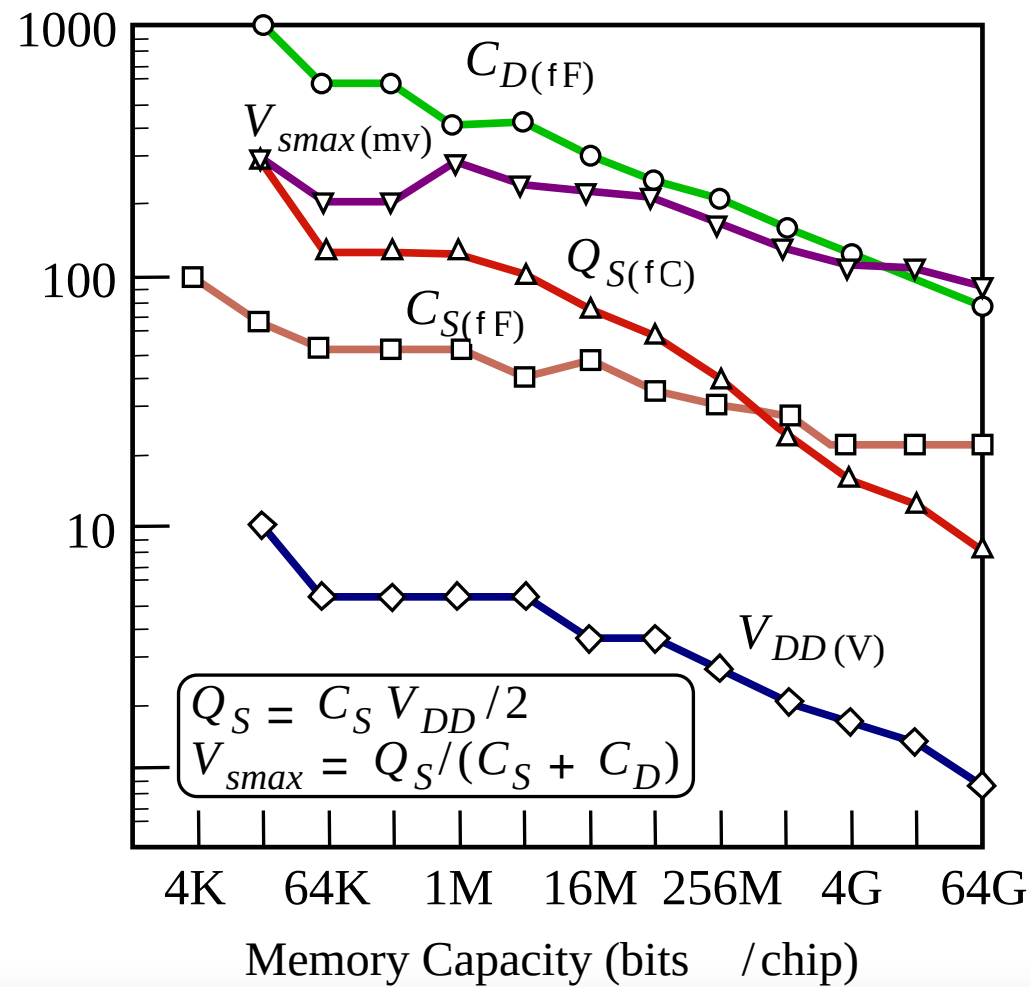
Equivalent Model



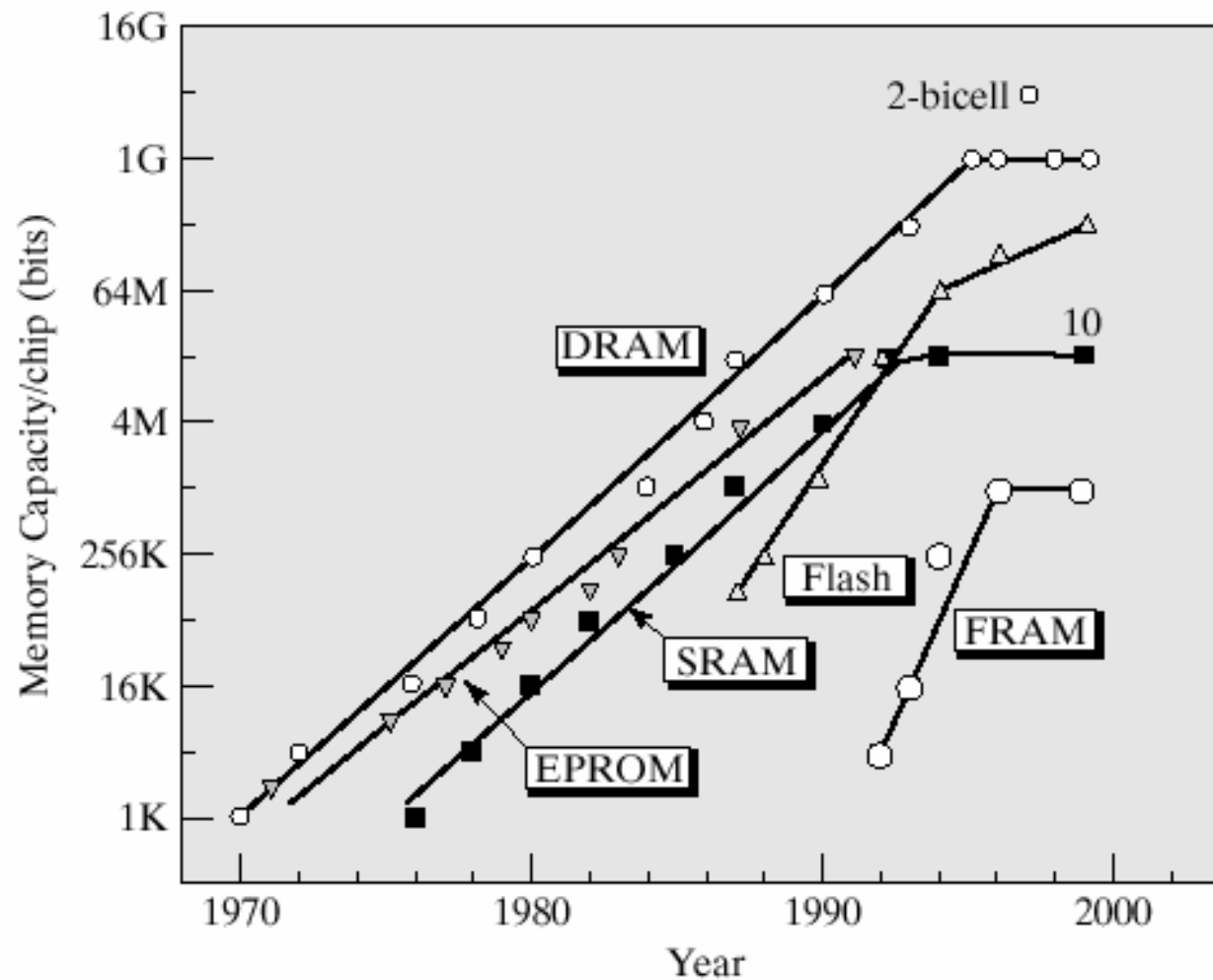
Charge Pump



Sensing-Parameters in DRAM



Semiconductor Memory Trends (updated)



Trends in Memory Cell Area

